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Item : 1 of 21

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Lab Values Notes Text Zoom

A wheelchair-bound 20-year-old man is brought to the emergency department because of fatigue and shortness of breath. Physical examination shows he has clusters of bright red blood vessels on the sclera of both eyes. A chest x-ray reveals a large mediastinal mass, which is subsequently identified as a non-Hodgkin lymphoma. Which of the following DNA repair mechanisms is most likely defective in this patient's cells?

- A. Base pair mismatch
- B. DNA polymerase proofreading
- C. Double strand break
- D. Nitrogenous base excision
- E. Nucleotide excision

Show Explanation

am200057m

A wheelchair-bound 20-year-old man is brought to the emergency department because of fatigue and shortness of breath. Physical examination shows he has clusters of bright red blood vessels on the sclera of both eyes. A chest x-ray reveals a large mediastinal mass, which is subsequently identified as a non-Hodgkin lymphoma. Which of the following DNA repair mechanisms is most likely defective in this patient's cells?

A. Base pair mismatch
B. DNA polymerase proofreading
C. **Double strand break**
D. Nitrogenous base excision
E. Nucleotide excision

Explanation & ReKaps

The correct answer is C. This patient possesses the traits associated with ***ataxia telangiectasia***, including ***telangiectasias of the skin and eyes***, variable immunodeficiency, and ***progressive ataxia***. Most patients are ***wheelchair bound*** by adolescence. The gene responsible for disease is called ***ATM*** and is classified as a ***tumor suppressor gene*** that encodes a ***DNA-dependent protein kinase*** localized mainly to the nucleus. This kinase is involved in the process that is responsible for recognizing ***double-strand DNA breaks*** and initiating the appropriate repair mechanisms. With the mutation of *ATM*, cells have ***increased sensitivity to ionizing radiation***, and exhibit defective DNA repair associated with frequent chromosomal abnormalities. Therefore, these individuals should avoid unnecessary x-ray exposures. The ***incidence of malignancy is high***, especially lymphoreticular malignancies ***(Hodgkin and non-Hodgkin lymphomas and leukemias)***.

ReKaps

- Ataxia telangiectasia is characterized by telangiectasias of the skin and eyes, variable immunodeficiency, and progressive ataxia.
- The ataxia telangiectasia gene (*ATM*) codes for a DNA-dependent protein kinase

Explanation & ReKaps

The correct answer is C. This patient possesses the traits associated with ***ataxia telangiectasia***, including ***telangiectasias of the skin and eyes***, variable immunodeficiency, and ***progressive ataxia***. Most patients are ***wheelchair bound*** by adolescence. The gene responsible for disease is called ***ATM*** and is classified as a ***tumor suppressor gene*** that encodes a ***DNA-dependent protein kinase*** localized mainly to the nucleus. This kinase is involved in the process that is responsible for recognizing ***double-strand DNA breaks*** and initiating the appropriate repair mechanisms. With the mutation of *ATM*, cells have ***increased sensitivity to ionizing radiation***, and exhibit defective DNA repair associated with frequent chromosomal abnormalities. Therefore, these individuals should avoid unnecessary x-ray exposures. The ***incidence of malignancy is high***, especially lymphoreticular malignancies ***(Hodgkin and non-Hodgkin lymphomas and leukemias)***.

Unresolved DNA replication errors can lead to mismatched base pairs **(choice A)**. These mismatches are normally identified by a protein complex that can recognize the bulge in the DNA molecule that is created by the mismatched bases. Once identified, the mismatch is then repaired by a separate complex with exonuclease activity. A mutation in one or more of these genes ultimately leads to an accumulation of DNA mutations due to the inability to eliminate the mismatches. ***Hereditary nonpolyposis colorectal cancer (Lynch syndrome)*** is a disease caused by mutations in genes involved in ***mismatch repair***, including ***MSH2*** and ***MLH1***. These individuals have an ***increased risk of colorectal cancers and endometrial cancers***.

During DNA replication, if the DNA polymerase inserts the incorrect nucleotide, it has the ability to correct the error. This process is known as DNA polymerase proofreading **(choice B)**, and is due to the polymerase's ***3' to 5' exonuclease activity***. Mutations that affect the proofreading function of DNA polymerase are associated with an ***increased risk of colorectal cancers***.

ReKaps

- Ataxia telangiectasia is characterized by telangiectasias of the skin and eyes, variable immunodeficiency, and progressive ataxia.
- The ataxia telangiectasia gene (*ATM*) codes for a DNA-dependent protein kinase responsible for recognizing and correcting DNA double strand breaks.
- Mutation of the *ATM* gene causes increased sensitivity to ionizing radiation and defective DNA repair associated with frequent chromosomal abnormalities.
- The incidence of lymphoreticular malignancies (Hodgkin and non-Hodgkin lymphomas and leukemias) is increased.

Once identified, the mismatch is then repaired by a separate complex with exonuclease activity. A mutation in one or more of these genes ultimately leads to an accumulation of DNA mutations due to the inability to eliminate the mismatches. ***Hereditary nonpolyposis colorectal cancer (Lynch syndrome)*** is a disease caused by mutations in genes involved in ***mismatch repair***, including ***MSH2*** and ***MLH1***. These individuals have an ***increased risk of colorectal cancers and endometrial cancers***.

During DNA replication, if the DNA polymerase inserts the incorrect nucleotide, it has the ability to correct the error. This process is known as DNA polymerase proofreading **(choice B)**, and is due to the polymerase's ***3' to 5' exonuclease activity***. Mutations that affect the proofreading function of DNA polymerase are associated with an ***increased risk of colorectal cancers***.

Defects in either nitrogenous base excision **(choice D)** or nucleotide base excision **(choice E)** can result in ***xeroderma pigmentosum***. This disease is characterized by ***extreme sensitivity to ultraviolet light***, resulting in ***excessive freckling, skin cancers, and corneal ulcerations***. Exposure to the sun results in ***thymine dimers*** along the DNA. Normally, the molecules involved with these repair mechanisms create nicks in the phosphodiester bonds flanking the thymine dimers, resulting in their removal. In xeroderma pigmentosum, mutations in the genes responsible for these activities (***XPA***, ***XPB***, ***XPC***, etc.) allow the thymine dimers to persist.

Updated on 12/07/15

- Mutation of the *ATM* gene causes increased sensitivity to ionizing radiation and defective DNA repair associated with frequent chromosomal abnormalities.
- The incidence of lymphoreticular malignancies (Hodgkin and non-Hodgkin lymphomas and leukemias) is increased.

Page References

MedEssentials (4th Ed.): pp. 64, 102

First Aid (2016): pp. 208.1-209.1

First Aid (2015): pp. 214.1-215.1

First Aid (2014): pp. 212.1-213.1

First Aid (2013): pp. 206.1-207.1

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Lab Values Notes Text Zoom

During an investigational study, cloning of a new eukaryotic gene is followed by insertion and ligation of the gene in an expression vector. The protein translated from this gene is then studied on a protein blot under non-denaturing conditions, and probed with ^{32}P-DNA. Which of the following techniques was most likely used in this study?

- A. Northern blot
- B. RNA interference
- C. Southwestern blot
- D. Western blot
- E. Yeast two hybrid

Show Explanation

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Lab Values Notes Text Zoom

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- A. Northern blot
- B. RNA interference
- C. **Southwestern blot**
- D. Western blot
- E. Yeast two hybrid

Explanation & ReKaps

The correct answer is C. The technique described is a ***Southwestern blot***, which is used to identify a ***DNA-protein interaction***. In this procedure, proteins in a sample are first separated by electrophoresis. These separated proteins are then transferred to a membrane and exposed to a ^{32}P-labelled probe composed of a specific DNA sequence. If a protein on the membrane is able to bind to that specific sequence, there will be a positive signal for that protein. ***Note that although both Northern blots and Southwestern blots use ^{32}P-DNA probes, only Southwestern blot can identify an interaction between a protein and DNA.***

Northern blotting **(choice A)** is helpful in determining which tissues express the gene for a specific protein. ***mRNA*** is isolated from samples of tissue separated by electrophoresis, blotted and probed with a ^{32}P-cDNA probe from the gene of interest. Results can reveal level of expression in different tissues or variability in length of the mRNAs, which may be result of alternative splicing. ***Northern blots are useful for analyzing abundance or size of mRNAs.***

ReKaps

DNA-binding proteins can be identified by ***Southwestern blotting***:

- First proteins are separated by electrophoresis, then blotted onto nitrocellulose paper.
- A ^{32}P labeled oligonucleotide produced by

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Explanation & ReKaps

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RNA interference**(choice B)** also known as RNAi technology can be used to reduce the expression of a specific gene, which allows for the study of gene function in either cell culture or whole model organisms. The nucleotide sequence of the interfering RNA molecule is complimentary to the sequence of the mRNA of interest. Once the RNAi molecule binds to the target mRNA, the mRNA becomes degraded, resulting in loss of the protein encoded by that mRNA. Any cellular effect due to loss of that protein can be identified. ***RNAi technology is useful for analyzing function(s) of specific gene(s) of interest.***

Western blotting **(choice D)** also called ***immunoblotting*** refers to transfer of proteins from a gel to a membrane and their detection using ***antibodies*** that specifically bind to the protein of interest. The antigen (protein)-antibody interaction enables ***the identification of a target protein***, as well as an ***estimation of its size***.

Yeast two hybrid**(choice E)** is a technique to detect ***protein-protein interactions that take place in cells***. In this technique, two molecules are introduced into a yeast cell: a protein of interest that is fused to one half of a transcription

ReKaps

DNA-binding proteins can be identified by ***Southwestern blotting***:

- First proteins are separated by electrophoresis, then blotted onto nitrocellulose paper.
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Page References

MedEssentials (4th Ed.): pp. 71

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Yeast two hybrid**(choice E)** is a technique to detect ***protein-protein interactions that take place in cells***. In this technique, two molecules are introduced into a yeast cell: a protein of interest that is fused to one half of a transcription factor, and a protein that is predicted to interact with the protein of interest, fused to the other half of a transcription factor. If the two proteins interact, a functional transcription factor is made, which results in the expression of genes that allows for growth under specific conditions. Following this process, the presence of colonies on selective media indicates an interaction between the protein of interest and the test protein.

Updated on 08/28/15

- First proteins are separated by electrophoresis, then blotted onto nitrocellulose paper.
- A ^{32}P-labeled oligonucleotide produced by restriction enzyme digestion of DNA (***^{32}P-DNA***) is used as a probe to detect protein-s DNA interactions.

Page References

MedEssentials (4th Ed.): pp. 71

First Aid (2016): pp. 65.1

First Aid (2015): pp. 78.1

First Aid (2014): pp. 82.1

First Aid (2013): pp. 80.1

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Lab Values Notes Text Zoom

A 7-year-old boy comes to the physician because of ptosis and drooping on the left side of his face for 2 days. He also has a 1-week history of pain and swelling in his right knee. His mother recalls that he had a tick bite several weeks ago that produced a ring-shaped rash. Which of the following techniques would most likely be appropriate for identifying seroconversion in this patient?

- A. Northern blotting
- B. Polymerase chain reaction (PCR)
- C. Restriction fragment length polymorphism (RFLP)
- D. Southern blotting
- E. Western blotting

Show Explanation

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Item : 3 of 21 Mark Previous Next Lab Values Notes Text Zoom

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A. Northern blotting
B. Polymerase chain reaction (PCR)
C. Restriction fragment length polymorphism (RFLP)
D. Southern blotting
E. **Western blotting**

Explanation & ReKaps

The correct answer is E. This patient demonstrates a late-phase presentation of ***Lyme disease***, which is caused by the ***spirochete Borrelia burgdorferi*** and is spread by the bite of infected ***Ixodes ticks***. *B. burgdorferi* is a large spirochete that is ***gram-negative*** and difficult to culture. Infected ticks are found on reservoirs such as white-tailed deer or white-footed mice; the spirochete is transferred to humans by the bite of the tick. *B. burgdorferi* invades through the skin and into the bloodstream; and ***affects the heart, joints, and the central nervous system (CNS)***.

Lyme disease initially presents as a ***rash, erythema migrans***, which is ***red and circular with a central clearing***. The rash resembles a ***bull's-eye***. Patients may also present with headache and general malaise during this phase. Weeks later, neurologic or cardiac symptoms may develop. Neurologically, patients may complain of a ***severe headache*** or ***Bell palsy*** (seen in the present case). Cardiac manifestations include arrhythmias and pericarditis. Months to years later, patients are likely to develop ***migratory arthritis***.

ReKaps

- Lyme disease is caused by the spirochete *Borrelia burgdorferi* that is transmitted by tick bite. A characteristic sign of this disease is the presence of a bull's eye rash at the site of the tick bite. Late stage disease is associated with neurologic and cardiac symptoms, as well as migratory arthritis.

Explanation & ReKaps

The correct answer is E. This patient demonstrates a late-phase presentation of ***Lyme disease***, which is caused by the ***spirochete Borrelia burgdorferi*** and is spread by the bite of infected ***Ixodes ticks***. *B. burgdorferi* is a large spirochete that is ***gram-negative*** and difficult to culture. Infected ticks are found on reservoirs such as white-tailed deer or white-footed mice; the spirochete is transferred to humans by the bite of the tick. *B. burgdorferi* invades through the skin and into the bloodstream; and ***affects the heart, joints, and the central nervous system (CNS)***.

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Upon infection, the individual's body will produce antibodies that specifically recognize the pathogen, and these ***antibodies*** will be ***present in the patient's blood***. This process is known as ***seroconversion***. The presence of these ***reactive antibodies*** can be detected by ***Western blotting***. This technique allows for the identification of multiple antibodies made against the spirochete, and is used to ***confirm the diagnosis***.

Northern blotting **(choice A)**, PCR **(choice B)**, RFLP **(choice C)**, and Southern blotting **(choice D)** are all methods that detect ***nucleic acids***, not proteins. Therefore, these would not be appropriate techniques for detecting the presence of antibodies in this patient's blood.

Updated on 10/06/15

ReKaps

- Lyme disease is caused by the spirochete *Borrelia burgdorferi* that is transmitted by tick bite. A characteristic sign of this disease is the presence of a bull's eye rash at the site of the tick bite. Late stage disease is associated with neurologic and cardiac symptoms, as well as migratory arthritis.
- Seroconversion refers to the presence of antibodies in an individual's blood that specifically recognize a pathogen with which the individual has been infected.
- Western blotting can be used to identify the presence of antibodies against *B. burgdorferi* in the patient's blood and confirm the diagnosis of an infection.

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Updated on 10/06/15

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Page References

MedEssentials (4th Ed.): pp. 71

First Aid (2016): pp. 65.1

First Aid (2015): pp. 78.1

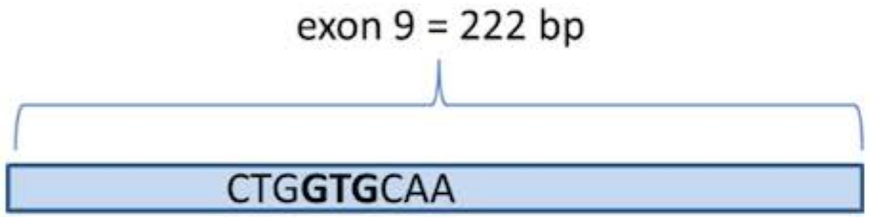
First Aid (2014): pp. 82.1

First Aid (2013): pp. 80.1

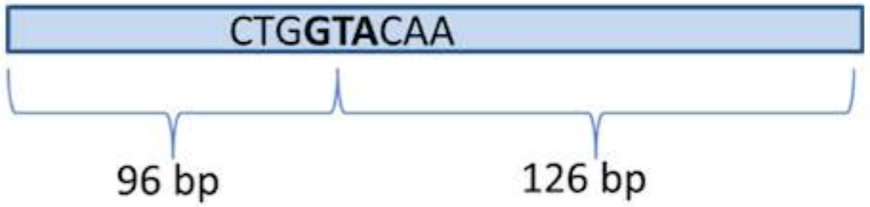
am200090m

An investigator uses RFLP analysis with Southern blotting to identify a point mutation in codon 408 of the LDL receptor gene after the amplification of exon 9 using PCR. This mutation results in the missense mutation of GTG (Val) to GTA (Met):

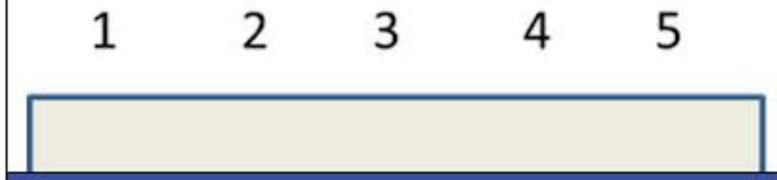
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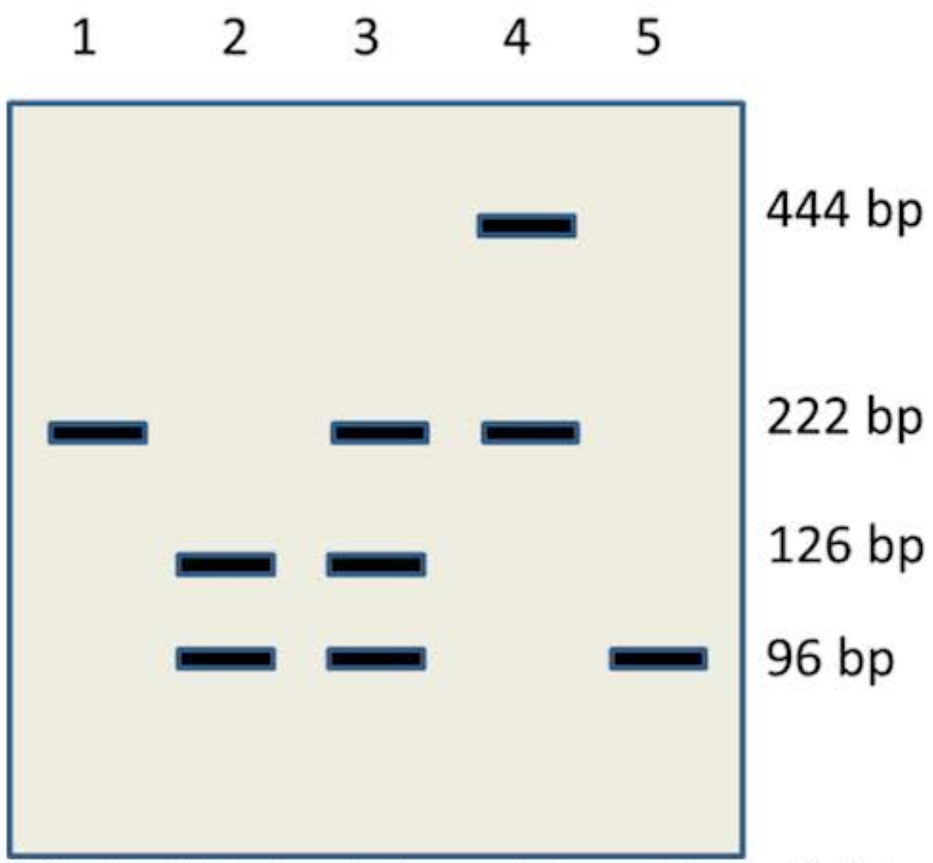
Mutant:



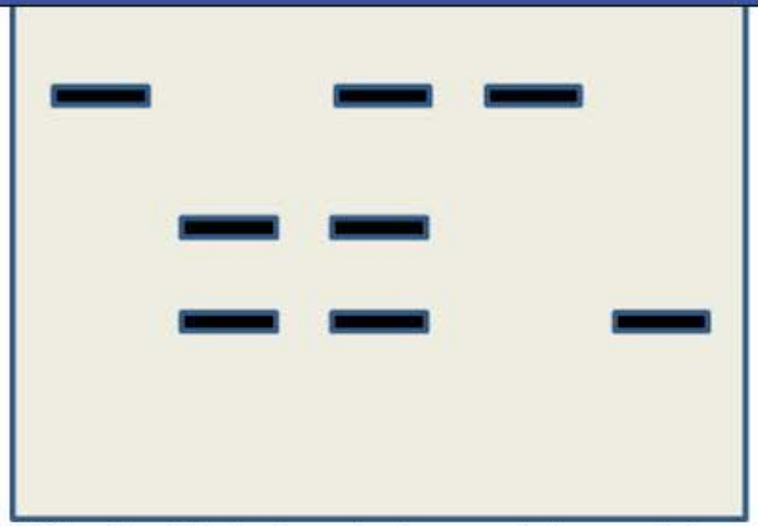
*Nla*III is a restriction enzyme that recognizes the short palindrome GTAC, which is not present in the normal exon 9. The amplified exon 9 is digested with this enzyme, and Southern blotting is performed with a probe directed against the entire amplified region.



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Which of the following lanes best represents the sample from a patient heterozygous for this mutation?

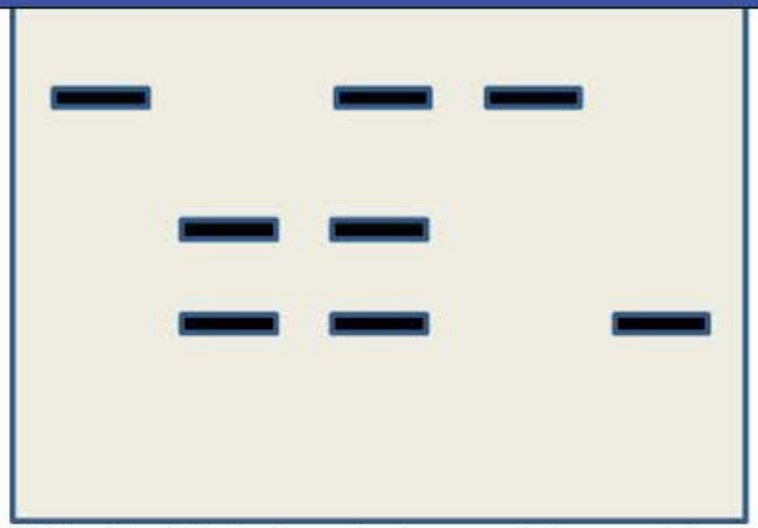


Which of the following lanes best represents the sample from a patient heterozygous for this mutation?

- A. Lane 1
- B. Lane 2
- C. Lane 3
- D. Lane 4
- E. Lane 5

Show Explanation

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Which of the following lanes best represents the sample from a patient heterozygous for this mutation?

- A. Lane 1
- B. Lane 2
- C. **Lane 3**
- D. Lane 4
- E. Lane 5

Explanation & ReKaps

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Explanation & ReKaps

The correct answer is C. ***R****estriction* ***F****ragment* ***L****ength* ***P****olymorphism* (**RFLP**) analysis determines a change in the genetic sequence that occurs at a site where a restriction enzyme cuts. The point mutation has resulted in the introduction of the palindrome recognized by *Nla*III, GTAC. ***Enzymatic digestion of the amplified 222 base-pair (bp) mutated exon 9 should therefore result in two fragments, of a combined mass of 222. Digestion of the normal 222-bp amplified exon should not result in fragmentation, and it should remain a 222-bp fragment.*** A heterozygote individual has one normal allele of the low-density lipoprotein (LDL) receptor gene and one mutated allele. Digestion of the polymerase chain reaction (PCR)-amplified exon 9 should therefore result in three different restriction fragments: the 222-bp normal allele and the fragmented mutant allele. Note that 126 + 96 = 222. Thus, Lane 3 is the correct answer.

Lane 1 (one band at 222) **(choice A)** would be expected in a ***homozygous normal*** individual, both alleles remaining intact after *Nla*III digestion.

Lane 2 (two bands of 126 and 96) **(choice B)** would be expected in a ***homozygous mutant*** individual, both alleles contain mutation and digested with *Nla*III enzyme.

Lane 4 (two bands of 444 and 222 bp) **(choice D)** would be very difficult to interpret. A fragment that is larger than the original amplified material could not be created by enzyme digestion.

Lane 5 (one band of 96 bp) **(choice E)** would be difficult to interpret. There should be another fragment of 126 and/or 222 bands.

Updated on 08/28/15

ReKaps

- PCR amplification and hybridization with allele specific oligonucleotides can easily and rapidly identify chromosomes that carry deletions or certain mutations. This technique is known as ***R****estriction* ***F****ragment* ***L****ength* ***P****olymorphism* (**RFLP**).
- If a mutation occurs at a site where a restriction enzyme cuts, ***RFLP*** analysis may be used determine the presence of a disease-causing allele of a gene.

Page References

MedEssentials (4th Ed.): pp. 69

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Updated on 08/28/15

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Page References

MedEssentials (4th Ed.): pp. 69

First Aid (2016): pp. 64.2

First Aid (2015): pp. 77.4

First Aid (2014): pp. 81.4

First Aid (2013): pp. 79.3

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Item : 5 of 21 Mark

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Lab Values Notes Text Zoom

A 48-year-old man has been recently diagnosed with colon cancer. His maternal uncle died at the age of 50 from colorectal carcinoma. Further studies show no evidence of familial polyposis but genetic testing shows a mutation in his MLH1 gene. Which of the following is most likely associated with this mutation?

- A. Active transcription of the telomerase gene
- B. Dinucleotide repeat instability
- C. Highly methylated CG sequences
- D. Increase in euchromatin
- E. Large segment deletions

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Item : 5 of 21 Mark Previous Next Lab Values Notes Text Zoom

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A. Active transcription of the telomerase gene
B. **Dinucleotide repeat instability**
C. Highly methylated CG sequences
D. Increase in euchromatin
E. Large segment deletions

Explanation & ReKaps

The correct answer is B. This man most likely has ***hereditary nonpolyposis colon cancer. Microsatellite instability*** is characteristic of individuals with loss-of-function mutations in the mismatch repair genes (***hMLH1, hMSH2***). These are associated with hereditary nonpolyposis colorectal cancer (HNPCC), hereditary endometrial cancer, and microsatellite instability in the tumor cells. Areas of DNA with dinucleotide repeats (an example of microsatellites) are often associated with ***strand slippage*** during DNA replication, allowing change in the number of repeats on the newly synthesized strand (microsatellite instability). The mismatch repair enzymes normally correct this replication error.

Active transcription of the telomerase gene **(choice A)** and increase in euchromatin **(choice D)** are both characteristics of many tumor cells, but are not known to be specifically related to loss of mismatch repair and HNPCC.

Highly methylated CG sequences **(choice C)** are typically associated with silencing certain gene regions; such methylation is not a prominent mechanism related to tumorigenesis

Page References

MedEssentials (4th Ed.): pp. 64
First Aid (2016): pp. 364.1
First Aid (2015): pp. 367.1
First Aid (2014): pp. 359.1
First Aid (2013): pp. 332.1

1 of 1 Playing
rachel-platten-lone-ranger (3:07)

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E. Large segment deletions

Explanation & ReKaps

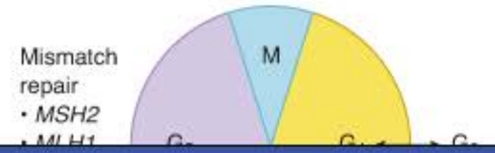
The correct answer is B. This man most likely has ***hereditary nonpolyposis colon cancer. Microsatellite instability*** is characteristic of individuals with loss-of-function mutations in the mismatch repair genes (***hMLH1, hMSH2***). These are associated with hereditary nonpolyposis colorectal cancer (HNPCC), hereditary endometrial cancer, and microsatellite instability in the tumor cells. Areas of DNA with dinucleotide repeats (an example of microsatellites) are often associated with ***strand slippage*** during DNA replication, allowing change in the number of repeats on the newly synthesized strand (microsatellite instability). The mismatch repair enzymes normally correct this replication error.

Active transcription of the telomerase gene **(choice A)** and increase in euchromatin **(choice D)** are both characteristics of many tumor cells, but are not known to be specifically related to loss of mismatch repair and HNPCC.

Highly methylated CG sequences **(choice C)** are typically associated with silencing certain gene regions; such methylation is not a prominent mechanism related to tumorigenesis.

Large segment deletions **(choice E)** are associated with unequal crossovers during meiosis. A large segment deletion might result in a genetic deficiency or loss of a tumor suppressor gene, and a hereditary form of cancer. The mechanism is not strongly associated with a defective mismatch repair system.

Important genes associated with maintaining fidelity of replicating DNA:



Page References

MedEssentials (4th Ed.): pp. 64

First Aid (2016): pp. 364.1

First Aid (2015): pp. 367.1

First Aid (2014): pp. 359.1

First Aid (2013): pp. 332.1

Choice A: 18% chose this answer.

Choice B: 47% chose this answer.

Choice C: 19% chose this answer.

Choice D: 6% chose this answer.

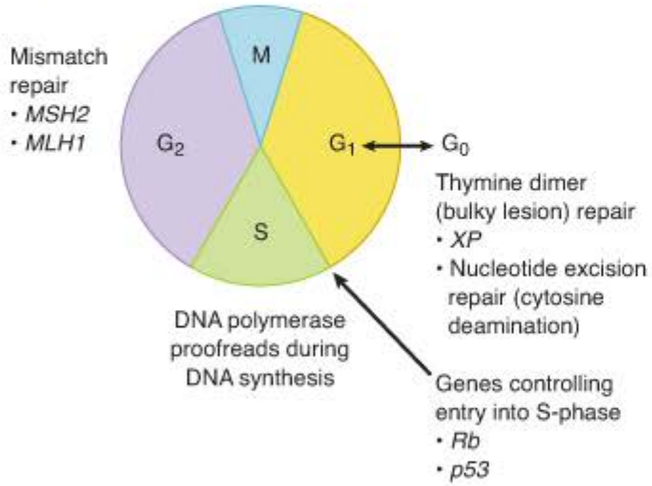
Choice E: 10% chose this answer.

of many tumor cells, but are not known to be specifically related to loss of mismatch repair and HNPCC.

Highly methylated CG sequences **(choice C)** are typically associated with silencing certain gene regions; such methylation is not a prominent mechanism related to tumorigenesis.

Large segment deletions **(choice E)** are associated with unequal crossovers during meiosis. A large segment deletion might result in a genetic deficiency or loss of a tumor suppressor gene, and a hereditary form of cancer. The mechanism is not strongly associated with a defective mismatch repair system.

Important genes associated with maintaining fidelity of replicating DNA:



Updated on 06/06/16

First Aid (2014): pp. 359.1

First Aid (2013): pp. 332.1

Choice A: 18% chose this answer.

Choice B: 47% chose this answer.

Choice C: 19% chose this answer.

Choice D: 6% chose this answer.

Choice E: 10% chose this answer.

aOL2066m

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A 1-month-old girl of northern European descent is brought to the physician for a follow-up examination because of productive cough, shortness of breath, and impaired growth. Her parents also report that she has been producing greasy, foul-smelling stools. The physician orders a sweat chloride test, which is positive. Which of the following best describes the mutation most likely responsible for this patient's condition?

- A. Frameshift
- B. Deletion
- C. Insertion
- D. Nonsense
- E. Substitution

Show Explanation

aOL2212m

A 1-month-old girl of northern European descent is brought to the physician for a follow-up examination because of productive cough, shortness of breath, and impaired growth. Her parents also report that she has been producing greasy, foul-smelling stools. The physician orders a sweat chloride test, which is positive. Which of the following best describes the mutation most likely responsible for this patient's condition?

A. Frameshift
B. **Deletion**
C. Insertion
D. Nonsense
E. Substitution

Explanation & ReKaps

The correct answer is B. This patient has the classic symptoms of ***cystic fibrosis***, including ***productive cough*** due to excess mucus, ***shortness of breath***, and ***greasy, foul-smelling stools***. Blockage of pancreatic ducts ***prevents the secretion of pancreatic enzymes*** necessary for the breakdown of fats in the diet, resulting in greasy, foul-smelling stools. This condition is termed ***pancreatic insufficiency***, and is common in severe cases of cystic fibrosis. The ***positive sweat chloride test*** is considered as the "gold standard" for diagnosing cystic fibrosis.

The most common mutation in individuals of northern European decent is the ***ΔF508mutation***. This mutation arises as a result of a deletion of three nucleotides in the coding region of the ***CFTR gene***, comprising the codon encoding phenylalanine (F) at position 508 of the amino acid sequence. The loss of this single amino acid ***prevents the proper folding of the protein***, resulting in its ***degradation***. Individuals who have this mutation in both of their *CFTR* alleles

ReKaps

- The deletion mutation ΔF508 is the most common mutation found in cystic fibrosis patients of northern European decent.
- Common symptoms of cystic fibrosis include productive cough due to excess mucus in the

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Explanation & ReKaps

The correct answer is B. This patient has the classic symptoms of ***cystic fibrosis***, including ***productive cough*** due to excess mucus, ***shortness of breath***, and ***greasy, foul-smelling stools***. Blockage of pancreatic ducts ***prevents the secretion of pancreatic enzymes*** necessary for the breakdown of fats in the diet, resulting in greasy, foul-smelling stools. This condition is termed ***pancreatic insufficiency***, and is common in severe cases of cystic fibrosis. The ***positive sweat chloride test*** is considered as the "gold standard" for diagnosing cystic fibrosis.

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Frameshift mutations **(choice A)** typically result from ***insertion or deletion of one or two base pairs in the coding region of a gene***. As a result, codons downstream from the mutation are "frame-shifted." Because three base pairs comprise a codon, insertions or deletions result in ***incorrect translation of all downstream codons***. Insertions and deletions have been reported in the *CFTR* gene that result in frameshift mutations, but these types of mutations are not common.

An insertion mutation **(choice C)** is created when ***one or more extra nucleotides are inserted into a gene***. Depending on the number of nucleotides inserted, this can result in a frameshift mutation or, if three nucleotides are inserted, the addition of a new amino acid to the protein encoded by the gene. Insertion mutations in cystic fibrosis patients have been identified, but they are not common.

Nonsense mutations **(choice D)** are mutations in which a ***stop codon is introduced*** upon the replacement of one

ReKaps

- The deletion mutation ΔF508 is the most common mutation found in cystic fibrosis patients of northern European decent.
- Common symptoms of cystic fibrosis include productive cough due to excess mucus in the lungs and greasy, foul smelling stools due to pancreatic insufficiency.
- A positive sweat chloride test is the "gold standard" for diagnosing cystic fibrosis.

Page References

MedEssentials (4th Ed.): pp. 64

phenylalanine (F) at position 508 of the amino acid sequence. The loss of this single amino acid ***prevents the proper folding of the protein***, resulting in its ***degradation***. Individuals who have this mutation in both of their *CFTR* alleles typically have a severe form of the disease.

Frameshift mutations **(choice A)** typically result from ***insertion or deletion of one or two base pairs in the coding region of a gene***. As a result, codons downstream from the mutation are "frame-shifted." Because three base pairs comprise a codon, insertions or deletions result in ***incorrect translation of all downstream codons***. Insertions and deletions have been reported in the *CFTR* gene that result in frameshift mutations, but these types of mutations are not common.

An insertion mutation **(choice C)** is created when ***one or more extra nucleotides are inserted into a gene***. Depending on the number of nucleotides inserted, this can result in a frameshift mutation or, if three nucleotides are inserted, the addition of a new amino acid to the protein encoded by the gene. Insertion mutations in cystic fibrosis patients have been identified, but they are not common.

Nonsense mutations **(choice D)** are mutations in which a ***stop codon is introduced*** upon the replacement of one nucleotide of a coding region with a different nucleotide. Some nonsense mutations, including G542X, cause cystic fibrosis, but they are not the most common mutation for this disease.

Substitution mutations **(choice E)** may occur as transversions or transitions. A ***transversion*** is the ***replacement of a purine with a pyrimidine***. A ***transition*** occurs when a ***purine is replaced by a different purine***. These types of mutations ***affect only one nucleotide*** and can result in either a missense or nonsense mutation if it occurs in the coding region of a gene. Some substitution mutations have been identified in cystic fibrosis patients, but these types of mutations are not as common as the ΔF508 deletion mutation.

Updated on 12/07/15

- Common symptoms of cystic fibrosis include productive cough due to excess mucus in the lungs and greasy, foul smelling stools due to pancreatic insufficiency.
- A positive sweat chloride test is the "gold standard" for diagnosing cystic fibrosis.

Page References

MedEssentials (4th Ed.): pp. 64

First Aid (2016): pp. 52.1

First Aid (2015): pp. 66.1

First Aid (2014): pp. 70.1

First Aid (2013): pp. 67.1

aOL2212m

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Lab Values Notes Text Zoom

An asymptomatic woman with a family history of β-thalassemia is pregnant with her second child. Genetic testing shows her husband is a carrier of a deletion of the β-globin gene ($\beta^n\beta^0$) and her first child is a carrier of a splice site mutation. Fetal DNA testing indicates that this child will likely have β-thalassemia major. Which of the following processes is most likely affected by this child's mutations?

- A. 5′ capping of mRNA
- B. β-globin gene rearrangements
- C. mRNA polyadenylation
- D. Removal of introns during RNA processing
- E. Translocation of mRNA to the cytoplasm

Show Explanation

am200058m

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A. 5′ capping of mRNA
B. β-globin gene rearrangements
C. mRNA polyadenylation
D. **Removal of introns during RNA processing**
E. Translocation of mRNA to the cytoplasm

Explanation & ReKaps

The correct answer is D. After messenger RNA (mRNA) has been produced (the primary transcript), it must undergo extensive posttranscriptional processing inside the nucleus to form the mature mRNA molecule. ***Removal of intervening sequences (introns)*** during processing of the primary transcript is carried out by ***spliceosomes (snRNP)***. The process requires that these structures ***recognize sequences at the 5′ and 3′ ends of the introns***. These sequences are referred to as ***5′ and 3′ splice sites***. A mutation in one of these splice sites would cause ***incorrect splicing***, and the β-globin produced would be less functional or nonfunctional.

An individual who has ***β-thalassemia major*** has ***mutations in both alleles*** of the β-thalassemia gene, resulting in ***defects in hemoglobin function***. Because this child's sibling has a mutation affecting the splice site in this gene, and the father only has a deletion mutation of this gene, it is very likely that the mother is a carrier of the splice site mutation.

ReKaps

- Splice site mutations would prevent the proper excision of introns because the recognition sites for spliceosomes would be altered.
- Splice site mutations would not affect the capping and poly-A additions occurring at the

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Explanation & ReKaps

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An individual who has ***β-thalassemia major*** has ***mutations in both alleles*** of the β-thalassemia gene, resulting in ***defects in hemoglobin function***. Because this child's sibling has a mutation affecting the splice site in this gene, and the father only has a deletion mutation of this gene, it is very likely that the mother is a carrier of the splice site mutation. Therefore, if the fetus is positive for β-thalassemia major, it must have inherited a deletion mutation from the father and a splice site mutation from the mother. The ***splice site mutation*** would lead to a ***defect in removal of introns from the primary transcript*** in this child's cells.

The β-globin gene is never rearranged **(choice B)**. ***Rearrangement of DNA*** occurs only in selected genes, including the ***genes for antibody heavy and light chains*** and the ***genes for the T-cell antigen receptors***.

Other ***posttranscriptional or cotranscriptional events*** include adding a ***5′ 7-methylguanosine triphosphate cap*** to the primary transcript **(choice A)** and adding a ***polyadenylate tail*** **(choice C)** to the 3′ end. Neither of these events involves splice sites.

Translocation of the fully processed mRNA to the cytoplasm **(choice E)** requires that intron removal and exon splicing has already been completed.

ReKaps

- Splice site mutations would prevent the proper excision of introns because the recognition sites for spliceosomes would be altered.
- Splice site mutations would not affect the capping and poly-A additions occurring at the 3′ and 5′ ends.
- β-thalassemia is a disease that affects the normal function of hemoglobin and can be caused by several different types of mutations, including deletions and splice site mutations.

Page References

Other ***posttranscriptional or cotranscriptional events*** include adding a ***5′ 7-methylguanosine triphosphate cap*** to the primary transcript **(choice A)** and adding a ***polyadenylate tail*** **(choice C)** to the 3′ end. Neither of these events involves splice sites.

Translocation of the fully processed mRNA to the cytoplasm **(choice E)** requires that intron removal and exon splicing has already been completed.

Updated on 05/09/16

Page References

MedEssentials (4th Ed.): pp. 65

First Aid (2016): pp. 54.4, 55.1

First Aid (2015): pp. 68.4, 69.1

First Aid (2014): pp. 73.2, 73.3

First Aid (2013): pp. 71.1, 71.2

Choice A: 4% chose this answer.

Choice B: 17% chose this answer.

Choice C: 14% chose this answer.

Choice D: 61% chose this answer.

Choice E: 4% chose this answer.

am200058m

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Item : 8 of 21

Mark

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Lab Values Notes Text Zoom

An 18-month-old girl is brought to the physician because the family noticed that in certain light, the pupil of her left eye appears white. Ophthalmologic examination shows a tumor in her left eye. She is diagnosed with retinoblastoma. Which of the following tests could be used to determine whether this is a heritable or sporadic tumor?

- A. ELISA
- B. Fluorescence-activated cell sorting (FACS)
- C. Northern blotting
- D. PCR
- E. Western blotting

Show Explanation

aOL331

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Item : 8 of 21 Mark Previous Next Lab Values Notes Text Zoom

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A. ELISA
B. Fluorescence-activated cell sorting (FACS)
C. Northern blotting
D. **PCR**
E. Western blotting

Explanation & ReKaps

The correct answer is D. One of the first signs of retinoblastoma is ***leukocoria***, which means "white pupil." It is most often initially noticed by close family members or on flash photography and is often the primary reason for consultation. Forty percent of retinoblastomas are ***heritable***; the child inherits one mutant allele through the germline and a somatic mutation in the other allele occurs at some point during retinal development, resulting in the loss of function of that gene. This leads to tumor development, ***often with multiple tumors in both eyes***. Children with heritable retinoblastoma usually are diagnosed within the first 5 years of life. Penetrance, however, is not complete because the second mutation is a chance event. Sixty percent of retinoblastomas are sporadic; both alleles are inactivated by somatic mutation within the same cell. Because this is a rare event, ***sporadic tumors usually are in only one eye and present at a later age***.

To detect a mutation in the ***Rb* gene** and differentiate between heritable and sporadic cases of retinoblastoma, ***PCR is the most appropriate technique*** of the choices given. Using the appropriate primers, ***a deletion can be detected as a***

ReKaps

- PCR is the test of choice to determine the sporadic or heritable nature of retinoblastoma.
- In heritable retinoblastoma, one chromosome will show a deletion or some other class of mutation in all the cells of the body.

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Explanation & ReKaps

The correct answer is D. One of the first signs of retinoblastoma is ***leukocoria***, which means "white pupil." It is most often initially noticed by close family members or on flash photography and is often the primary reason for consultation. Forty percent of retinoblastomas are ***heritable***; the child inherits one mutant allele through the germline and a somatic mutation in the other allele occurs at some point during retinal development, resulting in the loss of function of that gene. This leads to tumor development, ***often with multiple tumors in both eyes***. Children with heritable retinoblastoma usually are diagnosed within the first 5 years of life. Penetrance, however, is not complete because the second mutation is a chance event. Sixty percent of retinoblastomas are sporadic; both alleles are inactivated by somatic mutation within the same cell. Because this is a rare event, ***sporadic tumors usually are in only one eye and present at a later age***.

To detect a mutation in the ***Rb* gene** and differentiate between heritable and sporadic cases of retinoblastoma, ***PCR is the most appropriate technique*** of the choices given. Using the appropriate primers, ***a deletion can be detected as a change in size of the DNA band amplified***. In heritable cases, one chromosome will show the altered length from the deletion - in any cell in the body. In sporadic retinoblastoma, samples isolated from anywhere other than the tumor should have two normal alleles. Not all mutations in retinoblastoma are deletions. In these cases, PCR would be followed by testing the PCR products for other alleles known to cause retinoblastoma.

ELISA **(choice A)** is a highly sensitive assay that can be used to ***identify the presence of an antibody against a specific antigen or protein***. An example would be the ***ELISA HIV test***. In this test, antigens representing HIV viral proteins are fixed to the bottom of a plate, and a blood sample from an individual is added to this plate. If there are antibodies in the patient's blood, indicating an infection with the virus, a color change will indicate a positive result. ELISA cannot be used to distinguish between a sporadic or heritable form of cancer.

FACS **(choice B)** ***uses antibodies coupled to fluorescent markers to determine cell surface molecules on whole cells***. It is very useful for determining the stage of development or activation of cells (e.g., detection of CD antigens on

ReKaps

- PCR is the test of choice to determine the sporadic or heritable nature of retinoblastoma.
- In heritable retinoblastoma, one chromosome will show a deletion or some other class of mutation in all the cells of the body.
- In sporadic retinoblastoma, cells other than the tumor will show the normal two alleles.

Page References

MedEssentials (4th Ed.): pp. 69

First Aid (2016): pp. 64.2

First Aid (2015): pp. 77.4

have two normal alleles. Not all mutations in retinoblastoma are deletions. In these cases, PCR would be followed by testing the PCR products for other alleles known to cause retinoblastoma.

ELISA **(choice A)** is a highly sensitive assay that can be used to ***identify the presence of an antibody against a specific antigen or protein***. An example would be the ***ELISA HIV test***. In this test, antigens representing HIV viral proteins are fixed to the bottom of a plate, and a blood sample from an individual is added to this plate. If there are antibodies in the patient's blood, indicating an infection with the virus, a color change will indicate a positive result. ELISA cannot be used to distinguish between a sporadic or heritable form of cancer.

FACS **(choice B)** ***uses antibodies coupled to fluorescent markers to determine cell surface molecules on whole cells***. It is very useful for determining the stage of development or activation of cells (e.g., detection of CD antigens on cells in the immune system), but does not give you any insight into the DNA. This technique also allows isolation of cells with specific markers such as CD34+ stem cells.

Northern blotting **(choice C)**, the ***RNA counterpart to Southern blotting***, will determine the ***size and abundance of mRNA*** transcribed from a specific gene in a given sample of RNA. RNA is not able to be cleaved by restriction enzymes the way the DNA can; however, RNA molecules for different proteins are of different lengths and already short enough to be separated by gel electrophoresis. The sample is separated by agarose gel electrophoresis, transferred to a nitrocellulose membrane, and probed with a specific, labeled probe. This will be exposed to film and bands will appear, revealing the size and the abundance of specific mRNAs.

Western blotting **(choice E)** is the ***protein counterpart of Southern and Northern blotting***. Proteins are isolated from the cell type of interest and separated by size on a polyacrylamide gel, then transferred to a nitrocellulose membrane. The filter is then incubated with ***antibodies*** that specifically react with the protein of interest, then it is incubated with a labeled antibody (e.g., radioactive) that reacts with the first antibody. After this has been washed off, the membrane is exposed to film and bands characteristic of the protein being studied will appear. This is useful to determine whether a protein is present or absent in a patient or if it is the wrong size, suggesting an error in the DNA for this protein.

The following table compares inherited and sporadic retinoblastoma:

Page References

MedEssentials (4th Ed.): pp. 69

First Aid (2016): pp. 64.2

First Aid (2015): pp. 77.4

First Aid (2014): pp. 81.4

First Aid (2013): pp. 79.3

Choice A: 23% chose this answer.

Choice B: 8% chose this answer.

Choice C: 9% chose this answer.

Choice D: 50% chose this answer.

Choice E: 10% chose this answer.

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The following table compares inherited and sporadic retinoblastoma:

Inherited Retinoblastoma	Sporadic Retinoblastoma
Many tumors in both eyes	One tumor in one eye
Many tumors in other parts of the body	Tumors occur in other organs at the same rate as in the general population
Mutations are passed to offspring in autosomal dominant fashion	Mutations are not passed to offspring

Updated on 08/28/15

First Aid (2013): pp. 79.3

Choice A: 23% chose this answer.

Choice B: 8% chose this answer.

Choice C: 9% chose this answer.

Choice D: 50% chose this answer.

Choice E: 10% chose this answer.

aOL331

During an investigational study, cortisol is noted to increase gluconeogenesis in cultured hepatocytes. Which of the following processes is most directly involved in the observed cortisol-mediated stimulation of glucose production?

- A. Activation of adenylate cyclase
- B. Activation of a transmembrane tyrosine kinase
- C. Increased calcium influx from the extracellular fluid
- D. Increased transcription of gluconeogenic enzyme genes
- E. Termination of effects by phosphodiesterase

Show Explanation

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Item : 9 of 21 Mark

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Lab Values Notes Text Zoom

During an investigational study, cortisol is noted to increase gluconeogenesis in cultured hepatocytes. Which of the following processes is most directly involved in the observed cortisol-mediated stimulation of glucose production?

A. Activation of adenylate cyclase
B. Activation of a transmembrane tyrosine kinase
C. Increased calcium influx from the extracellular fluid
D. **Increased transcription of gluconeogenic enzyme genes**
E. Termination of effects by phosphodiesterase

Explanation & ReKaps

The correct answer is D. ***Steroid hormones diffuse across the plasma membrane and enter the cytoplasm, where they bind to an intracellular protein receptor.*** When a glucocorticoid, such as cortisol interacts with the cellular receptor, it undergoes a conformational change exposing a nuclear localization sequence. The hormone/receptor complex enters the nucleus and binds specific enhancer DNA sequences associated with certain genes, increasing mRNA transcription and thereby increasing protein synthesis. ***Cortisol*** increases ***gluconeogenic enzyme activity*** by increasing ***gene expression***, not by initiating intracellular signal transduction cascades.

Most peptide/protein hormones and paracrine factors bind to and activate specific plasma membrane receptors. The activated receptor initiates an intracellular transduction cascade that modifies the action of existing proteins and regulates the transcription of certain genes. Membrane receptors coupled through the heterotrimeric G_S to adenylate cyclase **(choice A)** or to phosphodiesterase **(choice E)** regulate intracellular levels of cyclic adenosine monophosphate (cAMP).

ReKaps

Cortisol induces key enzymes involved in gluconeogenesis by increasing gene expression.

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Explanation & ReKaps

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The receptors for ***insulin*** and for several other ***growth factors*** have tyrosine kinase activity that is activated by hormone binding **(choice B)**.

The NMDA receptor for the neurotransmitter glutamate includes a calcium channel activated by glutamate binding **(choice C)**.

Updated on 09/09/15

ReKaps

Cortisol induces key enzymes involved in gluconeogenesis by increasing gene expression.

Page References

MedEssentials (4th Ed.): pp. 169

First Aid (2016): pp. 316.2

First Aid (2015): pp. 321.3

First Aid (2014): pp. 316.1

First Aid (2013): pp. 295.1

Choice A: 17% chose this answer.

Choice B: 8% chose this answer.

gene expression, not by initiating intracellular signal transduction cascades.

Most peptide/protein hormones and paracrine factors bind to and activate specific plasma membrane receptors. The activated receptor initiates an intracellular transduction cascade that modifies the action of existing proteins and regulates the transcription of certain genes. Membrane receptors coupled through the heterotrimeric G_S to adenylate cyclase **(choice A)** or to phosphodiesterase **(choice E)** regulate intracellular levels of cyclic adenosine monophosphate (cAMP).

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Updated on 09/09/15

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Page References

MedEssentials (4th Ed.): pp. 169

First Aid (2016): pp. 316.2

First Aid (2015): pp. 321.3

First Aid (2014): pp. 316.1

First Aid (2013): pp. 295.1

Choice A: 17% chose this answer.

Choice B: 8% chose this answer.

Choice C: 3% chose this answer.

Choice D: 70% chose this answer.

Choice E: 2% chose this answer.

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A 3-year-old boy is tentatively diagnosed with X-linked chronic granulomatous disease. He has no immunoreactive cytochrome b_{245} beta chain. In order to determine whether the absence of this protein is due to failure of this gene to be transcribed or a post-transcriptional defect, mRNA is isolated from his neutrophils, subjected to gel electrophoresis, blotted, probed with a ^{32}P-labelled DNA probe specific for the cytochrome b_{245} beta chain gene, and visualized by autoradiography. Which of the following techniques best describes this procedure?

- A. Northern blotting
- B. Southern blotting
- C. Southwestern blotting
- D. Western blotting

Show Explanation

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A 3-year-old boy is tentatively diagnosed with X-linked chronic granulomatous disease. He has no immunoreactive cytochrome b_{245} beta chain. In order to determine whether the absence of this protein is due to failure of this gene to be transcribed or a post-transcriptional defect, mRNA is isolated from his neutrophils, subjected to gel electrophoresis, blotted, probed with a ^{32}P-labelled DNA probe specific for the cytochrome b_{245} beta chain gene, and visualized by autoradiography. Which of the following techniques best describes this procedure?

- A. **Northern blotting**
- B. Southern blotting
- C. Southwestern blotting
- D. Western blotting

Explanation & ReKaps

The correct answer is A. The technique described is ***Northern blotting***, an important research technique that has been ***very helpful in determining which tissues express the gene for a specific protein***. It gives a semi-quantitative result for the level of the gene expression in a tissue, and can reveal characteristics of gene expression such as alternative processing. The technique is performed as illustrated in the question stem.

Southern blotting **(choice B)** utilizes ***DNA-DNA hybridization***. The technique is similar to Northern blotting, but instead of RNA, DNA is extracted and analyzed. This procedure allows comparison of DNA samples for indirect genetic testing within families, and for relatedness of individuals. It can also be used to determine the epidemiologic relatedness of different bacterial biotypes, such as the strains of *Staphylococcus aureus* that produce toxic shock syndrome.

Southwestern blotting **(choice C)** involves ***DNA-protein interactions***. A protein sample extracted from cells, often from their nuclei, is subjected to electrophoresis, transferred to a filter, and exposed to labeled DNA. By examining the

ReKaps

- ***Northern blot*** is a technique used to determine whether a particular gene is expressed in a tissue or in several tissues.
- ***Southern blot*** can be used to compare DNA samples for genetic relatedness, and for indirect mutation analysis within families with a

Explanation & ReKaps

The correct answer is A. The technique described is ***Northern blotting***, an important research technique that has been ***very helpful in determining which tissues express the gene for a specific protein***. It gives a semi-quantitative result for the level of the gene expression in a tissue, and can reveal characteristics of gene expression such as alternative processing. The technique is performed as illustrated in the question stem.

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Southwestern blotting **(choice C)** involves ***DNA-protein interactions***. A protein sample extracted from cells, often from their nuclei, is subjected to electrophoresis, transferred to a filter, and exposed to labeled DNA. By examining the autoradiogram, one can identify proteins that bind to the labeled DNA and determine how large the DNA-binding protein(s) is/are. This can be used to identify transcription factors.

Western blotting **(choice D)** utilizes proteins identified by gel electrophoresis that are then transferred to a filter, where they can bind to ***specific antibodies used to identify the proteins***. One instance in which this technique is used is to separate false positive HIV-ELISA results from the true positives.

Blot Type	Material Analyzed	Probe used	Purpose
Southern	DNA	^{32}P-DNA	To determine which restriction fragments of DNA are associated with a particular gene
Northern	RNA	^{32}P-DNA	To measure sizes and

ReKaps

- ***Northern blot*** is a technique used to determine whether a particular gene is expressed in a tissue or in several tissues.
- ***Southern blot*** can be used to compare DNA samples for genetic relatedness, and for indirect mutation analysis within families with a genetic disease.
- ***Western blot*** is used to identify and locate proteins based on their ability to bind to specific antibodies.
- ***Southwestern blot*** is used to detect and characterize DNA-binding proteins and the specific motifs in the genomic DNA to which they bind.

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Blot Type	Material Analyzed	Probe used	Purpose
Southern	DNA	^{32}P-DNA	To determine which restriction fragments of DNA are associated with a particular gene
Northern	RNA	^{32}P-DNA	To measure sizes and amounts of specific mRNA molecules to answer questions about gene expression
Western	Protein	^{125}I- or enzyme-linked antibody	To measure amount of antigen (proteins) or antibody
Southwestern	Protein	^{32}P-DNA	To identify proteins that bind to DNA

Updated on 08/28/15

- ***Western blot*** is used to identify and locate proteins based on their ability to bind to specific antibodies.
- ***Southwestern blot*** is used to detect and characterize DNA-binding proteins and the specific motifs in the genomic DNA to which they bind.

Page References

MedEssentials (4th Ed.): pp. 71

First Aid (2016): pp. 65.1

First Aid (2015): pp. 78.1

First Aid (2014): pp. 82.1

First Aid (2013): pp. 80.1

Choice A: 71% chose this answer.

Choice B: 14% chose this answer.

Choice C: 7% chose this answer.

Choice D: 8% chose this answer.

			restriction fragments of DNA are associated with a particular gene
Northern	RNA	^{32}P-DNA	To measure sizes and amounts of specific mRNA molecules to answer questions about gene expression
Western	Protein	^{125}I- or enzyme-linked antibody	To measure amount of antigen (proteins) or antibody
Southwestern	Protein	^{32}P-DNA	To identify proteins that bind to DNA

Updated on 08/28/15

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Page References

MedEssentials (4th Ed.): pp. 71

First Aid (2016): pp. 65.1

First Aid (2015): pp. 78.1

First Aid (2014): pp. 82.1

First Aid (2013): pp. 80.1

Choice A: 71% chose this answer.

Choice B: 14% chose this answer.

Choice C: 7% chose this answer.

Choice D: 8% chose this answer.

Choice E: 0% chose this answer.

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Item : 11 of 21

Mark

Previous Next

Lab Values Notes Text Zoom

A newborn born at 39 weeks' gestation has several physical deformities, including thick gums and a large tongue. The newborn also has dislocation of the left hip, and clubbing of the feet, with relative immobility of the extremities. Fibroblast cultures from the infant show abnormal inclusions, and the culture medium contains lysosomal enzyme activity. This patient most likely suffers from defects in which of the following processes?

- A. Cleavage of N-terminal signal peptide
- B. Endocytosis
- C. Glycosylation of hydroxylysine
- D. Methylation of DNA
- E. Phosphorylation of mannose

Show Explanation

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Item : 11 of 21 Mark

Previous Next Lab Values Notes Text Zoom

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A. Cleavage of N-terminal signal peptide
B. Endocytosis
C. Glycosylation of hydroxylysine
D. Methylation of DNA
E. **Phosphorylation of mannose**

Explanation & ReKaps

The correct answer is E. The presence of ***lysosomal enzymes*** in the ***extracellular space*** is highly indicative of ***I-cell disease***, which is caused by ***deficiency*** in the enzyme UDP-N-acetylglucosamine: N -acetylglucosaminyl-1-phosphotransferase (***N-acetylglucosamine phosphotransferase***). This leads to ***decreased phosphorylation of carbon 6 in mannose residues*** in oligosaccharides on the N-linked ***glycoprotein acid hydrolases*** destined for the ***matrix of the lysosome***. For ***newly-translated*** lysosomal enzymes to be ***sorted*** through the ***Golgi apparatus*** and ***sent to lysosomes***, specific mannose residues must be phosphorylated at carbon 6. Deficiency of the phosphotransferase causes ***mistargeting*** of lysosomal enzymes to the ***exocytic pathway***, thereby leading to their ***release into the extracellular space***. As a result of mistargeting, ***undigested material (inclusions) accumulate*** in fibroblasts.

N-Terminal

ReKaps

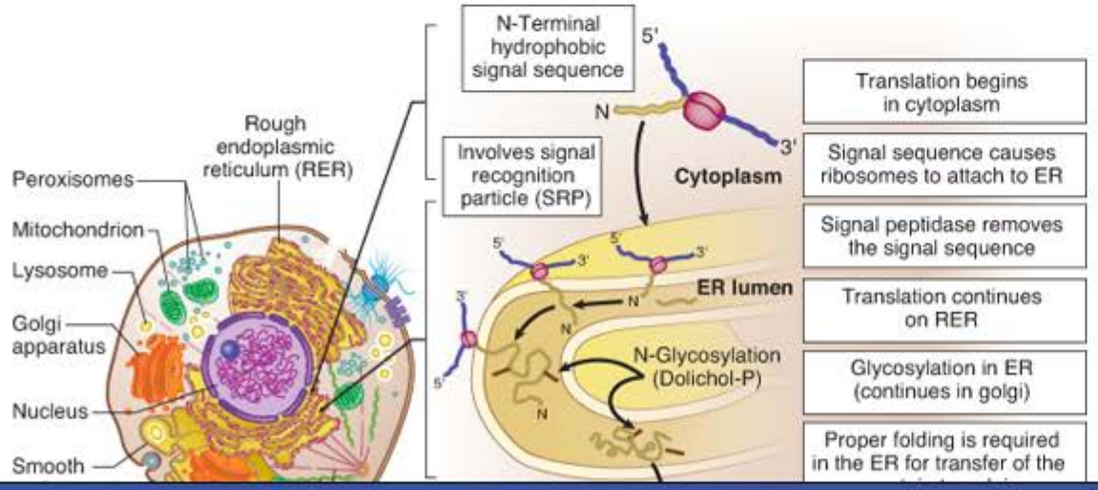
- I-cell disease is caused by a defect in the phosphotransferase that phosphorylates carbon 6 of selected mannose residues in the N-linked oligosaccharide on lysosomal enzymes. The lack of this mannose 6-phosphate lysosomal matrix-targeting signal

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Explanation & ReKaps

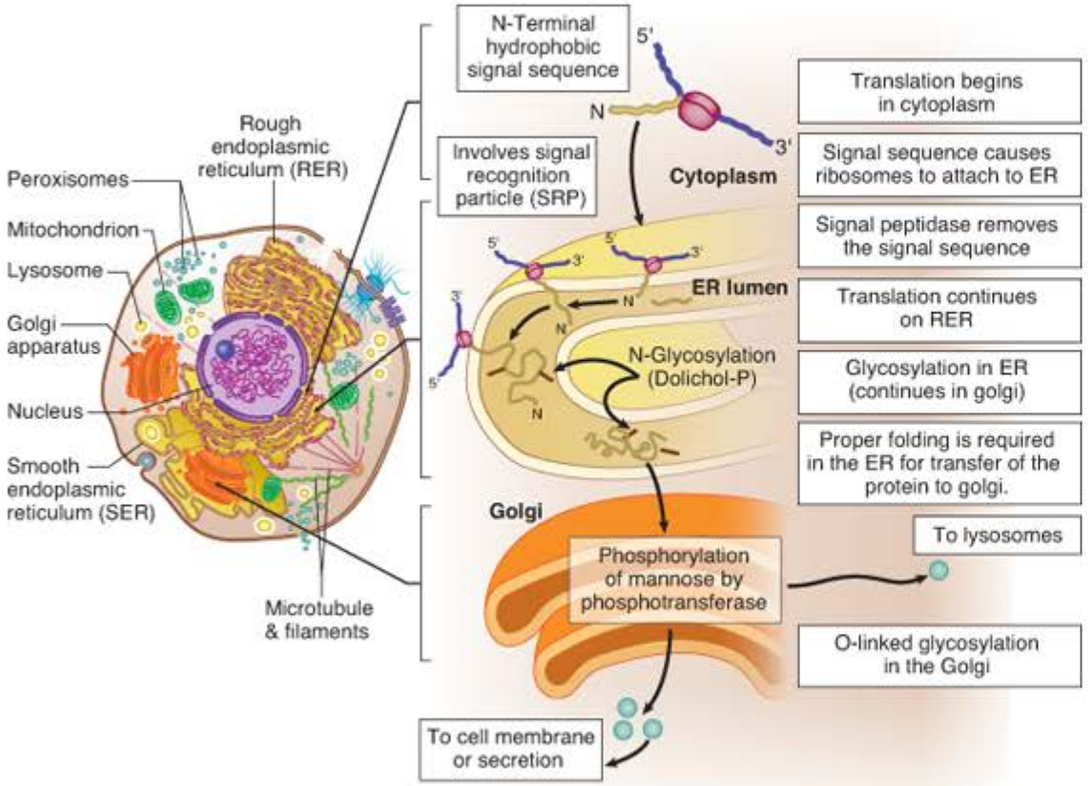
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ReKaps

- I-cell disease is caused by a defect in the phosphotransferase that phosphorylates carbon 6 of selected mannose residues in the N-linked oligosaccharide on lysosomal enzymes. The lack of this mannose 6-phosphate lysosomal matrix-targeting signal results in the enzymes being released into the extracellular space and undigested materials (intracellular inclusions) accumulating.
- Coarse facial features, skeletal abnormalities (such as congenital dislocation of the hip), and psychomotor abnormalities are characteristics of this mucolipidosis.

extracellular space. As a result of mistargeting, ***undigested material (inclusions) accumulate*** in fibroblasts.



I-cell disease is a ***rare autosomal recessive*** disorder in children that often presents with ***developmental delay and***

N-linked oligosaccharide on lysosomal enzymes. The lack of this mannose 6-phosphate lysosomal matrix-targeting signal results in the enzymes being released into the extracellular space and undigested materials (intracellular inclusions) accumulating.

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Page References

MedEssentials (4th Ed.): pp. 67

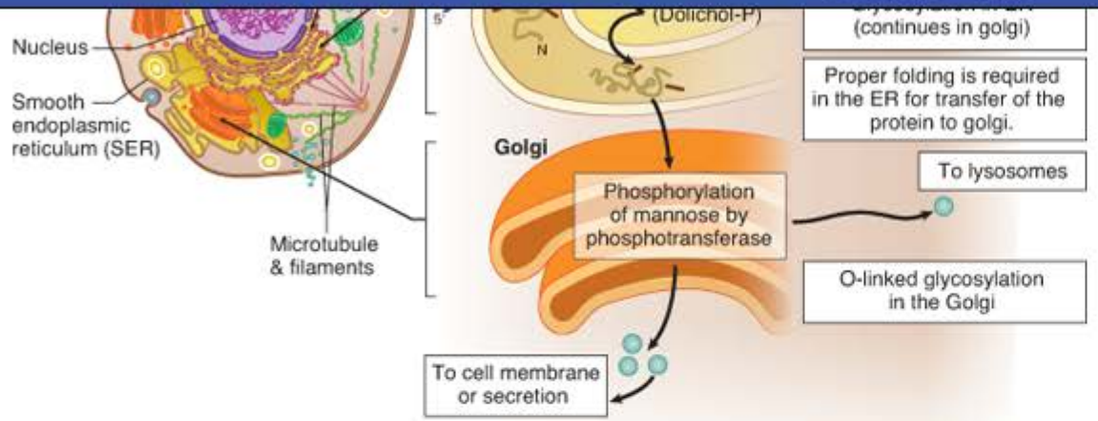
First Aid (2016): pp. 59.1

First Aid (2015): pp. 73.1

First Aid (2014): pp. 77.1

First Aid (2013): pp. 75.1

Choice A: 7% chose this answer.



I-cell disease is a ***rare autosomal recessive*** disorder in children that often presents with ***developmental delay and growth failure***. Linear growth often decelerates and ceases by age 2. Developmentally, patients often acquire a social smile and grasp objects, but they do not roll over and cannot support their weight to stand. They often have ***hypotonia*** and ***poor head control***. They have ***coarse facial features*** with gingival hypertrophy. Because the physical findings in I-cell disease overlap with those seen in some ***mucopolysaccharidoses and sphingolipidoses***, it is classified as a ***mucolipidosis (type II)***.

N-terminal signal peptide cleavage **(choice A)** occurs in the ***lumen of the rough endoplasmic reticulum (ER)***. It is a ***required step*** in the ***cotranslational targeting of proteins*** to the ***interior*** of the ***ER***, ***Golgi,*** and ***lysosomes***; ***most membranes***; and the ***extracellular environment***. ***Mutations*** in this process are most likely ***embryonic-lethal***, as numerous proteins rely on its function.

In endocytosis **(choice B)** a portion of the plasma membrane invaginates at ***clathrin-coated pits*** and pinches off forming a membrane-bounded vesicle called an ***endosome***. There are three types of endocytosis: ***phagocytosis, pinocytosis,***

Page References

MedEssentials (4th Ed.): pp. 67

First Aid (2016): pp. 59.1

First Aid (2015): pp. 73.1

First Aid (2014): pp. 77.1

First Aid (2013): pp. 75.1

Choice A: 7% chose this answer.

Choice B: 5% chose this answer.

Choice C: 16% chose this answer.

Choice D: 5% chose this answer.

Choice E: 67% chose this answer.

smile and grasp objects, but they do not roll over and cannot support their weight to stand. They often have ***hypotonia*** and ***poor head control***. They have ***coarse facial features*** with gingival hypertrophy. Because the physical findings in I-cell disease overlap with those seen in some ***mucopolysaccharidoses and sphingolipidoses***, it is classified as a ***mucolipidosis (type II)***.

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In endocytosis **(choice B)** a portion of the plasma membrane invaginates at ***clathrin-coated pits*** and pinches off forming a membrane-bounded vesicle called an ***endosome***. There are three types of endocytosis: ***phagocytosis, pinocytosis, and receptor-mediated endocytosis***. The latter is seen with uptake of the ***low-density lipoprotein (LDL) receptor*** and the ***transferrin receptor***. If endocytosis is negatively affected it would lead to the ***inability*** of cells to ***take up extracellular material.***

Glycosylation of hydroxylysine **(choice C)** is an early ***post-translational modification*** of the ***α chains*** that form the ***triple helical collagen protein***. While the exact function of the O-glycosylation is unknown, ***mutations*** in this intracellular process would most likely lead to ***skin and musculoskeletal abnormalities***.

DNA methylation **(choice D)** is an ***epigenetic modification to DNA*** that most commonly results in ***gene silencing***. ***Normal methylation*** is seen in ***X-inactivation and genomic imprinting***. ***Hypermethylation*** has been shown to occur in patients with ***fragile X syndrome*** and, in combination with the effects of trinucleotide repeat expansion, leads to various ***mental and physical abnormalities***.

Updated on 08/31/15

Choice C: 16% chose this answer.

Choice D: 5% chose this answer.

Choice E: 67% chose this answer.

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Item : 12 of 21 Mark

Previous Next

Lab Values Notes Text Zoom

A 29-year-old woman requests a paternity test to determine if her current boyfriend is the father of her child. Blood samples are taken from the patient, the son, and the patient's boyfriend. Which of the following procedures would be most appropriate for analyzing these samples?

- A. Chromosomal karyotyping
- B. Microsatellite analysis
- C. Northern blot analysis
- D. Promoter methylation
- E. Western blot analysis

Show Explanation

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Item : 12 of 21 Mark Previous Next Lab Values Notes Text Zoom

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- A. Chromosomal karyotyping
- B. **Microsatellite analysis**
- C. Northern blot analysis
- D. Promoter methylation
- E. Western blot analysis

Explanation & ReKaps

The correct answer is B. *Microsatellite testing detects genetic differences between individuals* due to ***different lengths of microsatellite repeats***. Microsatellites are repeated units of typically 2-4 bp, and the length of the entire repeat is generally less than 150 bp. These ***simple tandem repeats (STRs)*** create a ***molecular fingerprint*** of an individual, as no two persons would have the same pattern of STRs. Using this technique, it is possible to identify the parents of a child due to the presence of STR bands in the child's sample that represent STR bands from each of the parents.

In case 1 below, the tested male may be the father because he shares a band with the child. It is not certain, however, because other men in the population could have the same band. Matches are required at 9 to 10 different loci to indicate with high probability that a tested male is the father. In case 2, the tested male cannot be the father because neither of his bands is shared with the child.

ReKaps

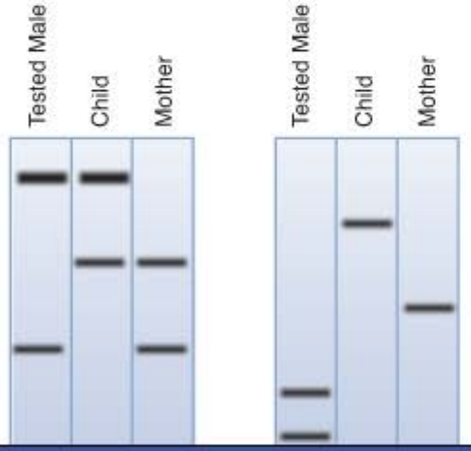
- Paternity testing is most often performed by utilizing PCR probes that amplify microsatellite repeats. A series of 4 to 8 different probes is generally used to confirm the paternity.
- The pattern of microsatellite repeats is compared between the mother, the child, and

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Explanation & ReKaps

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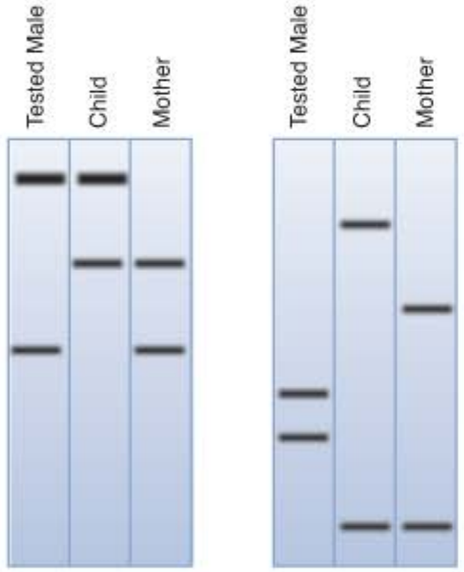
ReKaps

- Paternity testing is most often performed by utilizing PCR probes that amplify microsatellite repeats. A series of 4 to 8 different probes is generally used to confirm the paternity.
- The pattern of microsatellite repeats is compared between the mother, the child, and the tested male.
- If the tested male is truly the child's biological father, at least 1 of the child's band lengths will match 1 of those of the father.

Page References

MedEssentials (4th Ed.): pp. 71

because other men in the population could have the same band. Matches are required at 9 to 10 different loci to indicate with high probability that a tested male is the father. In case 2, the tested male cannot be the father because neither of his bands is shared with the child.



CASE 1 **CASE 2**

Chromosomal karyotyping **(choice A)** ***detects chromosomal abnormalities*** such as trisomies or monosomies. It can be used to ***diagnose various genetic diseases***, but it cannot be used to determine paternity.

Northern blot analysis **(choice C)** can be used to determine the expression pattern or expression levels of specific ***RNA molecules***. It is commonly used to ***compare mRNA levels*** among different tissue types. RNA levels or expression patterns cannot be used to identify a particular individual or that individual's progeny.

Promoter methylation analysis **(choice D)** is used to identify ***epigenetic silencing*** of genes. ***Methylation of cytosines*** in

- The pattern of microsatellite repeats is compared between the mother, the child, and the tested male.
- If the tested male is truly the child's biological father, at least 1 of the child's band lengths will match 1 of those of the father.

Page References

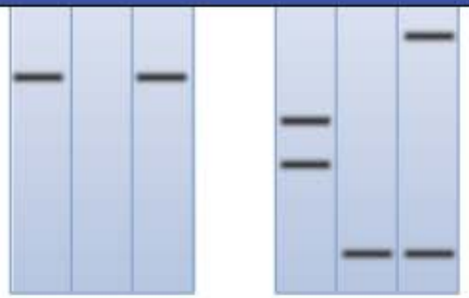
MedEssentials (4th Ed.): pp. 71

First Aid (2016): pp. 64.2

First Aid (2015): pp. 77.4

First Aid (2014): pp. 81.4

First Aid (2013): pp. 79.3



CASE 1 CASE 2

Chromosomal karyotyping **(choice A)*detects chromosomal abnormalities*** such as trisomies or monosomies. It can be used to ***diagnose various genetic diseases***, but it cannot be used to determine paternity.

Northern blot analysis **(choice C)** can be used to determine the expression pattern or expression levels of specific ***RNA molecules***. It is commonly used to ***compare mRNA levels*** among different tissue types. RNA levels or expression patterns cannot be used to identify a particular individual or that individual's progeny.

Promoter methylation analysis **(choice D)** is used to identify ***epigenetic silencing*** of genes. ***Methylation of cytosines*** in promoters can prevent the expression of the associated gene, which is a mechanism used by some cancers to ***inhibit the expression of tumor suppressors***. This method cannot be used to determine the paternity of a child.

Western blot analysis **(choice E)** allows for the ***detection of specific proteins*** in a sample. This type of analysis can used to determine the presence/absence of a protein, the size of the protein, or the level of expression of a protein. Protein size/levels would not allow one to determine the paternity of a child.

Updated on 05/24/16

Page References

MedEssentials (4th Ed.): pp. 71

First Aid (2016): pp. 64.2

First Aid (2015): pp. 77.4

First Aid (2014): pp. 81.4

First Aid (2013): pp. 79.3

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Item : 13 of 21 Mark

Previous Next Lab Values Notes Text Zoom

A 46-year-old man is brought to the physician by his wife because of a 6-month history of progressive memory loss and personality changes. His wife says that his father developed similar symptoms at the age of 60. Physical examination shows involuntary twitching movements of the limbs and poor coordination. Which of the following genetic mechanisms is most likely involved in this disorder?

- A. Expanded trinucleotide tandem repeat
- B. Genomic imprinting
- C. Large deletion in one gene
- D. Single amino acid substitution
- E. Translocation

Show Explanation

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Item : 13 of 21 Mark Previous Next Lab Values Notes Text Zoom

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- A. **Expanded trinucleotide tandem repeat**
- B. Genomic imprinting
- C. Large deletion in one gene
- D. Single amino acid substitution
- E. Translocation

Explanation & ReKaps

The correct answer is A. Several disorders, including ***Huntington disease***, are caused by the expansion of a three-base sequence within a specific gene on chromosome 4. The majority of ***trinucleotide expansion disorders*** share certain common features. Normal individuals possess a ***range of copy numbers*** for the pathogenic trinucleotide. These normal copy numbers are relatively stable over meiosis, so that individuals in the next generation inherit the trinucleotide copy number as a simple, stable, Mendelian trait. However, some individuals display premutation, a characteristic in which the copy number of the trinucleotide is increased and unstable but their phenotype is normal. Individuals with premutations may pass on further expanded trinucleotide tracts to their offspring.

The presence of ***more trinucleotide repeats in the offspring***, resulting in an ***earlier age of onset*** and a ***more severe phenotype***, is a phenomenon known as ***anticipation***. Because this patient is presenting with symptoms much earlier in

ReKaps

- Huntington disease is an example of a disease caused by a trinucleotide expansion within the coding region of a gene.
- In Huntington disease, the expansion of CAG repeats is thought to produce a toxic "gain of

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Explanation & ReKaps

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The presence of ***more trinucleotide repeats in the offspring***, resulting in an ***earlier age of onset*** and a ***more severe phenotype***, is a phenomenon known as ***anticipation***. Because this patient is presenting with symptoms much earlier in life than his father, this is most likely a case of anticipation. The mechanism by which trinucleotide repeats cause disease varies. In Huntington disease, the ***expansion of CAG repeats*** is thought to produce a toxic "gain of function." This means that the disease develops because the mutant form of the protein gains a new or greater function. Other genetic diseases caused by trinucleotide expansion include fragile X syndrome, myotonic dystrophy, and Friedreich's ataxia.

Prader-Willi and ***Angelman syndromes*** are frequently cited examples of genetic diseases involving ***genomic imprinting*** **(choice B)**. On the ***maternal chromosome***, the genes responsible for ***Prader-Willi syndrome*** are normally silenced by an epigenetic mechanism. Similarly, on the ***paternal chromosome***, the ***Angelman syndrome*** genes are epigenetically silenced. This produces a situation in which an individual is functionally heterozygous for these alleles. If a deletion occurs in the Prader-Willi genes on the paternal chromosome, then the individual has lost his only active copies of those genes, and he will have Prader-Willi. If a deletion occurs in the Angelman syndrome genes on the maternal chromosome, then that individual will no longer have any active Angelman genes, and he will have Angelman syndrome.

ReKaps

- Huntington disease is an example of a disease caused by a trinucleotide expansion within the coding region of a gene.
- In Huntington disease, the expansion of CAG repeats is thought to produce a toxic "gain of function."
- An earlier age of onset and increased severity of the disease in offspring due to the inheritance of extra trinucleotide repeats is known as anticipation.

Page References

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Prader-Willi and ***Angelman syndromes*** are frequently cited examples of genetic diseases involving ***genomic imprinting*** **(choice B)**. On the ***maternal chromosome***, the genes responsible for ***Prader-Willi syndrome*** are normally silenced by an epigenetic mechanism. Similarly, on the ***paternal chromosome***, the ***Angelman syndrome*** genes are epigenetically silenced. This produces a situation in which an individual is functionally heterozygous for these alleles. If a deletion occurs in the Prader-Willi genes on the paternal chromosome, then the individual has lost his only active copies of those genes, and he will have Prader-Willi. If a deletion occurs in the Angelman syndrome genes on the maternal chromosome, then that individual will no longer have any active Angelman genes, and he will have Angelman syndrome.

Duchenne muscular dystrophy is an X-linked recessive disease due to a large ***deletion*** **(choice C)** in the gene that encodes ***dystrophin***. This deletion creates a ***frameshift*** in the coding region of the gene, preventing the formation of functional dystrophin protein. Large deletions occur in 60-70% of cases and small point mutations occur in 15-30% of cases.

Single amino acid substitutions **(choice D)** are common in recessive diseases. An example is ***sickle cell anemia***, in which ***glutamic acid is replaced with valine*** in position 6 of the protein's amino acid sequence.

Translocations **(choice E)** of chromosomes give rise to several disorders, including ***chronic myelogenous leukemia*** (CML; Philadelphia chromosome), in which a translocation occurs between chromosomes 9 and 22, ***t (9;22)***, producing the oncogenic fusion protein ***BCR-ABL***.

Updated on 06/06/16

known as anticipation.

Page References

MedEssentials (4th Ed.): pp. 72, 216

First Aid (2016): pp. 439.2-69.2, 71.1, 73.3

First Aid (2015): pp. 80.1-81.2, 83.1, 85.3

First Aid (2014): pp. 84.1-85.1, 87.1, 89.3

First Aid (2013): pp. 82.1, 85.1, 87.2

Choice A: 94% chose this answer.

Choice B: 2% chose this answer.

Choice C: 2% chose this answer.

Choice D: 1% chose this answer.

Choice E: 1% chose this answer.

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Item : 14 of 21

Mark

Previous Next

Lab Values Notes Text Zoom

A 5-year-old boy is brought to the physician because of pain in his left arm and right leg. Physical examination shows blue-tinged sclera, hearing loss, and small, slightly blue, misshapen teeth. An x-ray of the left arm shows a comminuted fracture of the humerus and an x-ray of the right leg shows a comminuted fracture of the femur. These fractures are in different stages of healing. Which of the following are components of the molecule most likely defective in this patient?

- A. α/β-tubulin dimers
- B. Fibrillin
- C. Hydroxylated lysines and prolines
- D. Keratin sulfate
- E. Repeating disaccharide units

Show Explanation

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Item : 14 of 21 Mark

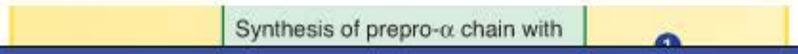
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A. α/β-tubulin dimers
B. Fibrillin
C. **Hydroxylated lysines and prolines**
D. Keratin sulfate
E. Repeating disaccharide units

Explanation & ReKaps

The correct answer is C. The suspected disease is ***osteogenesis imperfecta***, which is a rare genetic disorder that occurs in both recessive and dominant forms. The clinical presentation, depending on the specific form, varies from death *in utero*, to that described in the question stem, to very mild disease with only a modest increase in bone fragility. The different types all are characterized by ***defects in the synthesis of type I collagen***, which is a ***major component of bone***, contributing to its rigidity and strength. Inside the cell, type I collagen undergoes post-translation modifications during its synthesis, with ***hydroxylation of prolines and lysines*** being the first modification (see figure). Allelic heterogeneity within the *ColA1* gene or *ColA2* gene produces an unstable collagen triple helix that is not as strong as normal collagen, but phenotypic outcomes depend on the specific mutation(s) in either of these genes.



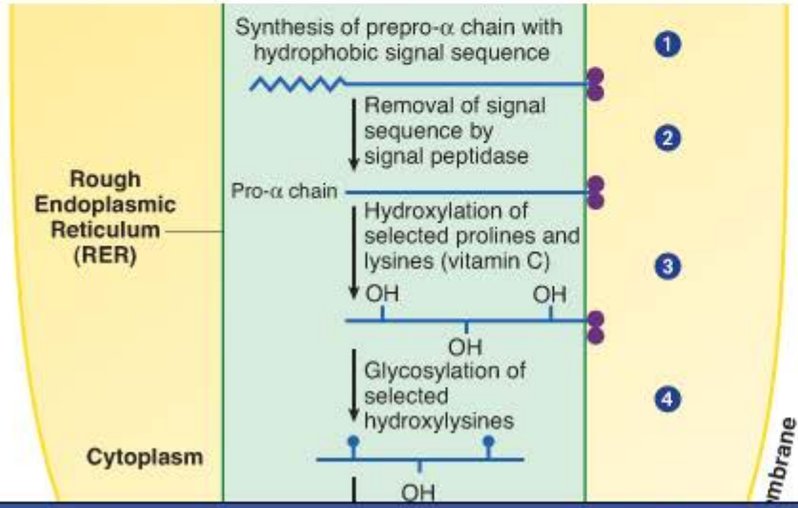
ReKaps

- Osteogenesis imperfecta (OI) is an inherited connective-tissue disorder with many phenotypic presentations.
- Most patients with OI have mutations that affect the structure of one of the two chains of

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Explanation & ReKaps

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ReKaps

- Osteogenesis imperfecta (OI) is an inherited connective-tissue disorder with many phenotypic presentations.
- Most patients with OI have mutations that affect the structure of one of the two chains of type I collagen, which is hydroxylated on prolines and lysines prior to further post-translational modifications.

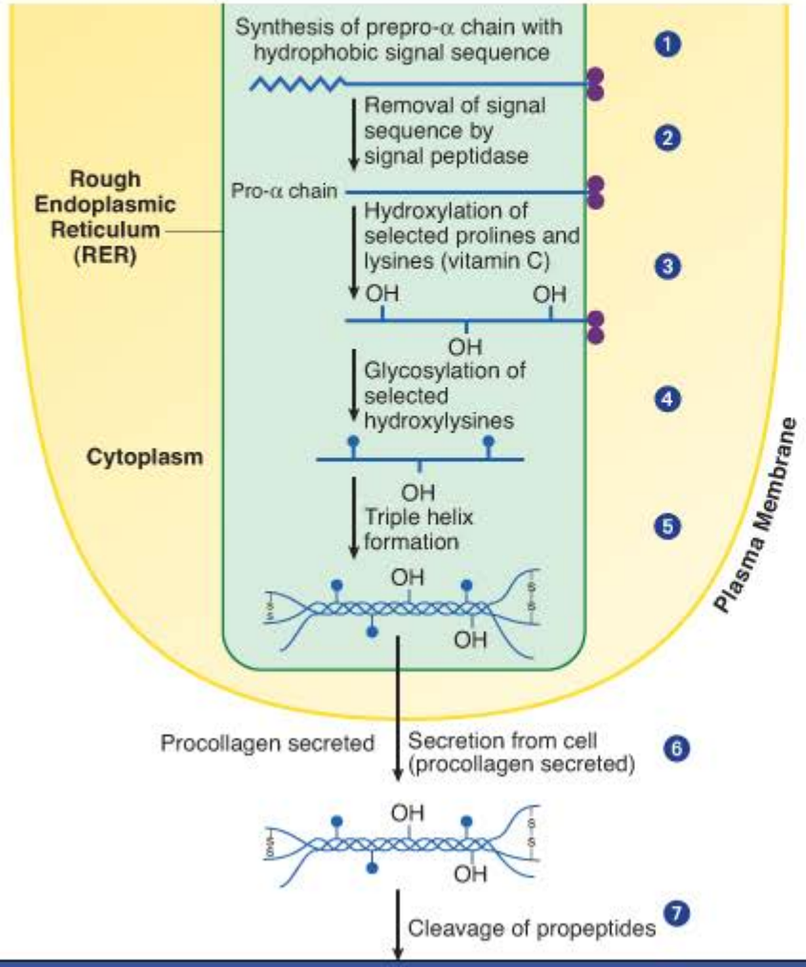
Page References

MedEssentials (4th Ed.): pp. 68, 442

First Aid (2016): pp. 63.1

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Page References

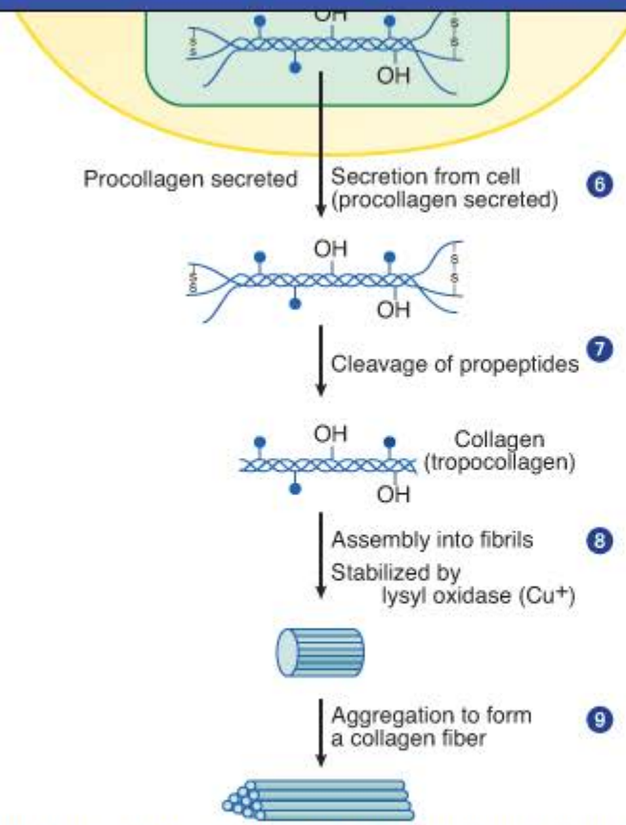
MedEssentials (4th Ed.): pp. 68, 442

First Aid (2016): pp. 63.1

First Aid (2015): pp. 76.2

First Aid (2014): pp. 80.2

First Aid (2013): pp. 78.2

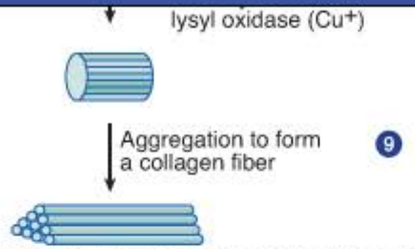


First Aid (2014): pp. 80.2

First Aid (2013): pp. 78.2

α/β-tubulin dimers **(choice A)** are ***subunits of microtubules***. Microtubules, along with actin microfilaments and intermediate filaments, make up the ***cytoskeleton*** inside the cell. They function to maintain the shape and integrity of the cell, but they do not contribute to bone strength. The type I collagen found in the bone matrix is mainly responsible for this property.

Fibrillin **(choice B)** is a protein that ***polymerizes to form microfibrils***. The ***microfibrils combine with elastin to form***



α/β-tubulin dimers **(choice A)** are ***subunits of microtubules***. Microtubules, along with actin microfilaments and intermediate filaments, make up the ***cytoskeleton*** inside the cell. They function to maintain the shape and integrity of the cell, but they do not contribute to bone strength. The type I collagen found in the bone matrix is mainly responsible for this property.

Fibrillin **(choice B)** is a protein that ***polymerizes to form microfibrils***. The ***microfibrils combine with elastin to form elastic fibers***, an extracellular matrix component. These elastic fibers allow the connective tissue of various tissues, including blood vessels, lungs, and skin, to stretch. ***Marfan syndrome*** is caused by a ***defect in elastin fibers due to a mutation in the gene encoding fibrillin-1***.

Keratan sulfate **(choice D)** is a sulfated glycosaminoglycan that is attached to a core protein, creating a ***proteoglycan***. This proteoglycan can assemble with other proteoglycans and GAGs to produce a large molecule called ***aggrecan***. Aggrecan is a ***major component of cartilage*** and ***provides resistance to compressive forces***.

Repeating disaccharide units **(choice E)** describes a ***glycosaminoglycan (GAG)*** molecule. GAGs contain ***unbranched polysaccharide chains composed of repeating disaccharide units***. GAGs are a ***major component of cartilage matrix*** and ***provide resistance to compression*** due to their strong hydrophilic properties.

Updated on 10/07/15

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Item : 15 of 21

Mark

Previous Next

Lab Values Notes Text Zoom

A 2-week-old girl is brought to the physician because of vomiting and a yellow appearance. Her temperature is 37°C (98.6°F), pulse is 138/min, and respirations are 29/min. Physical examination shows a lethargic infant with jaundice, hepatomegaly, and a poor sucking reflex. Serum studies show an elevated total bilirubin level. Northern blot analysis shows a normal amount of galactose-1-phosphate uridyltransferase (GALT) mRNA, but biochemical studies do not show observable enzyme activity. Which of the following is the best explanation for this patient's condition?

- A. Gene deletion
- B. Nonsense mutation
- C. Premature transcription termination sequence in the DNA
- D. Promoter mutation
- E. RNA splicing mutation

Show Explanation

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- A. Gene deletion
- B. **Nonsense mutation**
- C. Premature transcription termination sequence in the DNA
- D. Promoter mutation
- E. RNA splicing mutation

Explanation & ReKaps

The correct answer is B. ***Galactose-1-phosphate uridyltransferase (GALT)*** is a key enzyme in the metabolism of galactose. Lack of a functional enzyme causes ***galactosemia***, an autosomal recessive condition characterized by vomiting, hyperbilirubinemia with jaundice, hepatomegaly, and cataracts. Northern blot is a technique that studies gene expression by analyzing mRNA. A nonsense mutation is a change in one DNA base pair that leads to a premature stop codon when the mRNA is being translated into protein. The mRNA would be transcribed correctly, but when the protein was being translated, it would be stopped prematurely, leading to a truncated protein, which functions abnormally.

Immediate minimization of dietary galactose is essential. Human milk and formula based on bovine milk should be discontinued and replaced with soy formula. This will decrease the acute toxicity but does not prevent all long-term complications of this disease, and monitoring of these patients throughout their lives is necessary. Patients with this

ReKaps

- Nonsense mutations generate one of the three stop codons.
- Transcription proceeds, but the truncated translation product may be grossly abnormal in shape and function or may be so unstable

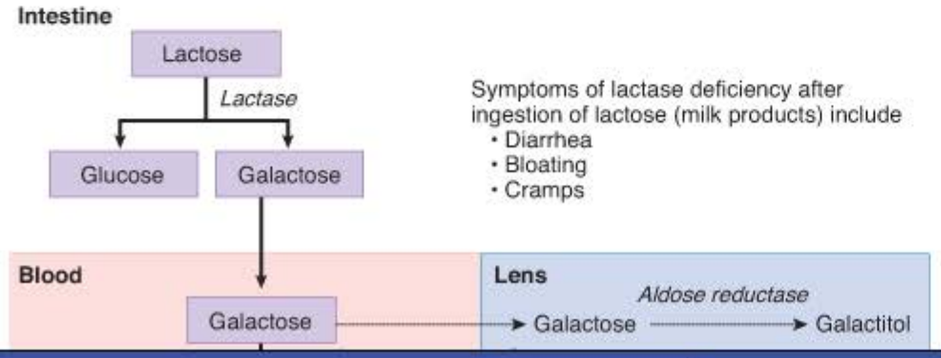
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Explanation & ReKaps

The correct answer is B. *Galactose-1-phosphate uridyltransferase (GALT)* is a key enzyme in the metabolism of galactose. Lack of a functional enzyme causes ***galactosemia***, an autosomal recessive condition characterized by vomiting, hyperbilirubinemia with jaundice, hepatomegaly, and cataracts. Northern blot is a technique that studies gene expression by analyzing mRNA. A nonsense mutation is a change in one DNA base pair that leads to a premature stop codon when the mRNA is being translated into protein. The mRNA would be transcribed correctly, but when the protein was being translated, it would be stopped prematurely, leading to a truncated protein, which functions abnormally.

Immediate minimization of dietary galactose is essential. Human milk and formula based on bovine milk should be discontinued and replaced with soy formula. This will decrease the acute toxicity but does not prevent all long-term complications of this disease, and monitoring of these patients throughout their lives is necessary. Patients with this disease require a lactose restricted diet for life.

A review of galactose metabolism is provided below.



ReKaps

- Nonsense mutations generate one of the three stop codons.
- Transcription proceeds, but the truncated translation product may be grossly abnormal in shape and function or may be so unstable that it is rapidly degraded.

Page References

MedEssentials (4th Ed.): pp. 38, 64

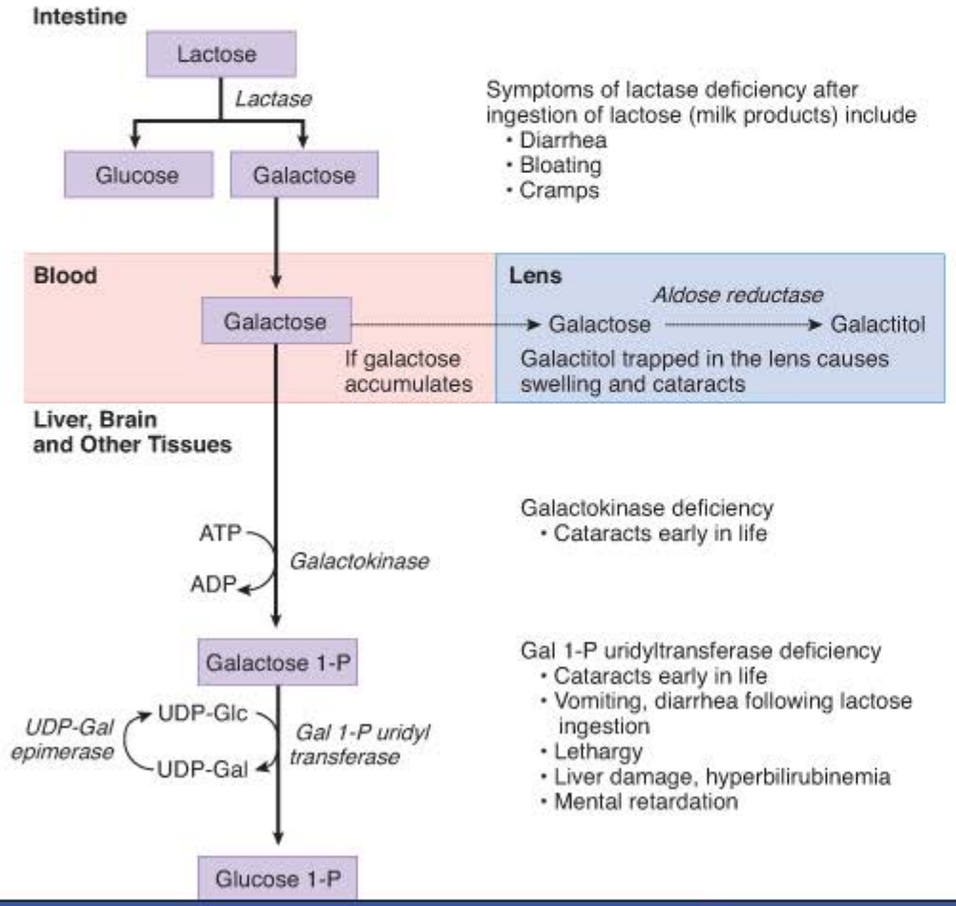
First Aid (2016): pp. 52.1, 91.2

First Aid (2015): pp. 66.1, 103.2

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Page References

MedEssentials (4th Ed.): pp. 38, 64

First Aid (2016): pp. 52.1, 91.2

First Aid (2015): pp. 66.1, 103.2

First Aid (2014): pp. 70.1, 107.1

First Aid (2013): pp. 67.1, 105.1

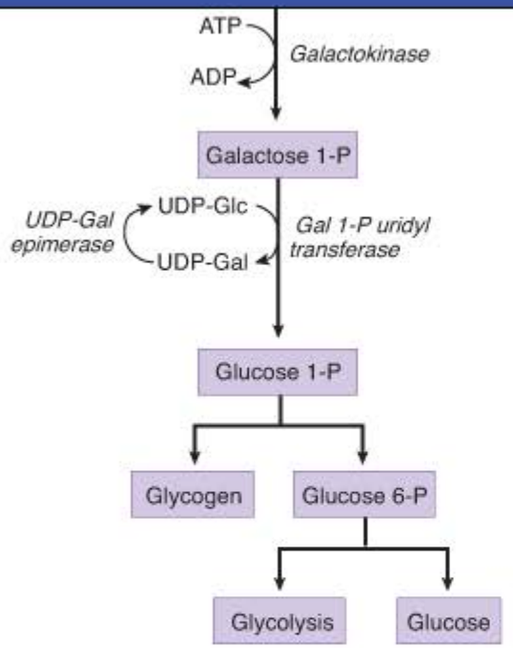
Choice A: 4% chose this answer.

Choice B: 31% chose this answer.

Choice C: 11% chose this answer.

Choice D: 15% chose this answer.

Choice E: 39% chose this answer.



• Cataracts early in life

Gal 1-P uridyltransferase deficiency
- Cataracts early in life
- Vomiting, diarrhea following lactose ingestion
- Lethargy
- Liver damage, hyperbilirubinemia
- Mental retardation

In the well-fed state, galactose can enter glycolysis or contribute to glycogen storage

Administration of galactose during hypoglycemia induces an increase in blood glucose

A gene deletion **(choice A)** would result in either the absence of mRNA or mRNA of a different size, both of which would be apparent by Northern blot analysis.

A premature transcription termination sequence in the DNA **(choice C)** might result in premature termination of the mRNA chain, leading to a diminished or absent message.

A promoter mutation **(choice D)** in the DNA, if it caused the gene in question to be less actively transcribed, would result in a diminished amount of message.

An RNA splicing mutation **(choice E)** would result in the production of an mRNA molecule of a different size, which would be apparent on Northern blot.

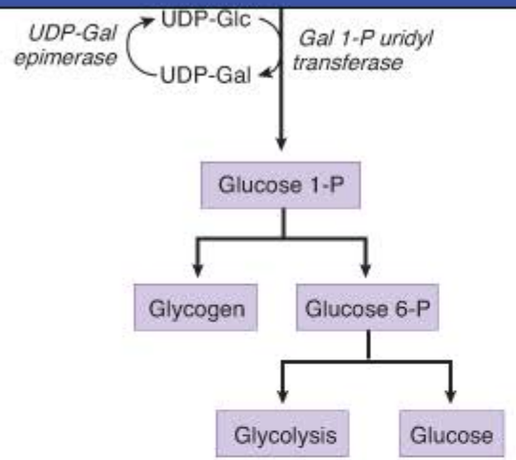
Choice B: 31% chose this answer.

Choice C: 11% chose this answer.

Choice D: 15% chose this answer.

Choice E: 39% chose this answer.

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- Liver damage, hyperbilirubinemia
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Choice E: 39% chose this answer.

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Updated on 04/18/16

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Item : 16 of 21

Mark

Previous Next

Lab Values Notes Text Zoom

A 50-year-old woman comes to the physician because she found a lump in her right breast during self examination. She has is unaware of any family history of cancer because she was adopted, has been otherwise healthy, and has not had a mammogram. Her temperature is 37.2°C (99°F), pulse is 72/min, respirations are 14/min, and blood pressure is 130/76 mm Hg. Physical examination shows a 2-cm mass in the right breast as well as firm, non-tender axillary lymph nodes. The left breast is normal. A mammogram reveals a stellate dense lesion with focal calcifications. Histological examination of a biopsy sample identifies the tumor as an invasive ductal carcinoma with positive HER-2 immunohistochemical staining. This tumor is most likely to have high activity of which of the following proteins?

- A. Gαs
- B. GLI
- C. Ras
- D. SMAD
- E. STAT

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Item : 16 of 21 Mark Previous Next Lab Values Notes Text Zoom

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- A. Gαs
- B. GLI
- C. **Ras**
- D. SMAD
- E. STAT

Explanation & ReKaps

The correct answer is C. ***HER-2***, also known as ***ERBB2***, is a growth factor receptor found on the surface of cells. It is a ***receptor tyrosine kinase*** (RTK) and is a member of the ***epidermal growth factor receptor (EGFR) family***. ***Amplification of HER-2*** is found in up to 30% of ***sporadic breast cancers*** and is, thus, an important marker for these tumors. In healthy tissue, the activity of these receptors is necessary for ductal development in the mammary glands. Growth factor receptors such as HER-2 become activated by binding of their ligand. This results in dimerization of the receptor, followed by autophosphorylation of its intracellular tails. The activated receptor leads to ***activation of Ras***, which in turn activates members of the ***MAPKKK*** family, ultimately leading to the expression of genes necessary to induce

ReKaps

- HER-2 is a member of the epidermal growth factor receptor family and signals through Ras and MAP kinases.

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Explanation & ReKaps

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Gαs **(choice A)** is a component of some forms of the ***heterotrimeric G protein complex***. Other members of this complex include the ***β and γ subunits***. The G protein complex associates with ***G protein coupled receptors (GPCRs)***. For some GPCR family members, binding of ligand to the receptor leads to the activation of the Gαs subunit. Gαs, in turn, activates ***adenylyl cyclase***, resulting in production of ***cAMP***. Mutations in GCPRs are not associated with breast cancer.

GLI **(choice B)** activity is important in normal embryonic development. It is induced by binding of the signaling molecule ***SHH*** to the ***PTCH1 receptor***. GLI is an example of an ***oncogene***, as mutations resulting in increased activity of the protein are found in many types of tumors, including ***glioblastoma***. Mutations in ***PTCH1***, resulting in high levels of GLI-1 activity, are associated with ***basal cell carcinoma***.

SMAD **(choice D)** functions to regulate gene expression as a downstream component of the ***TGFβ signaling pathway***. Mutations or deletions in this gene are important contributors to some forms of cancer, such as ***pancreatic***

ReKaps

- HER-2 is a member of the epidermal growth factor receptor family and signals through Ras and MAP kinases.
- Many breast cancers have HER-2 amplification, resulting in high levels of Ras signaling, resulting in over-proliferation of the cells.

Page References

MedEssentials (4th Ed.): pp. 84, 160

First Aid (2016): pp. 230.1

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SMAD **(choice D)** functions to regulate gene expression as a downstream component of the ***TGFβ signaling pathway***. Mutations or deletions in this gene are important contributors to some forms of cancer, such as ***pancreatic adenocarcinoma***, in which more than half (55%) of all cases have lost SMAD activity.

High levels ofSTAT protein **(choice E)** activity give rise to some forms of ***myeloproliferative neoplasms***. In these diseases, ***JAK,*** a ***tyrosine kinase*** upstream of STAT, is mutated to a constitutively active form, resulting in high levels of STAT activity downstream and increased proliferation of the cell.

Updated on 12/01/15

- Many breast cancers have HER-2 amplification, resulting in high levels of Ras signaling, resulting in over-proliferation of the cells.

Page References

MedEssentials (4th Ed.): pp. 84, 160

First Aid (2016): pp. 230.1

First Aid (2015): pp. 236.1

First Aid (2014): pp. 232.1

First Aid (2013): pp. 221.1

Pathoma (2015): pp. 24.4, 26 table

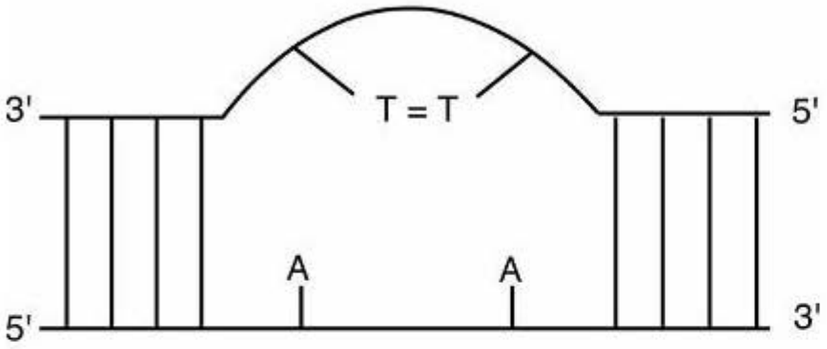
Pathoma (2014): pp. 24.4, 26 table

Pathoma (2013): pp. 24.4, 26, table

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Item : 17 of 21 Mark



A study is conducted to characterize a familial condition in which affected individuals develop severe sunburn upon exposure to small amounts of sunlight. Examination of DNA from these individuals reveals a large number of mutations such as the one shown in the diagram. Which of the following is most likely to develop in individuals with this genetic disease?

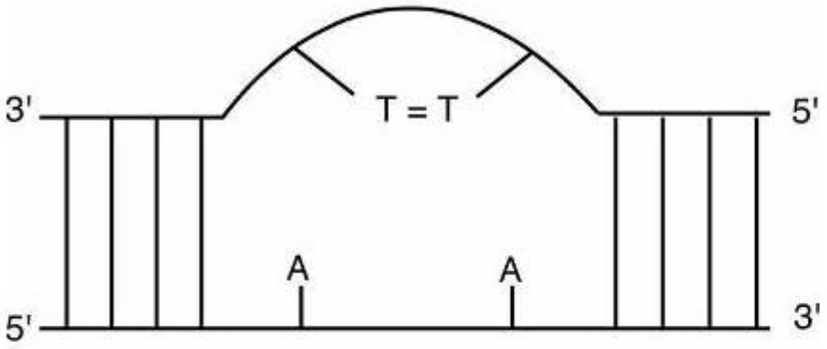
- A. Basal cell carcinoma
- B. Colorectal adenocarcinoma
- C. Ductal carcinoma
- D. Hepatocellular carcinoma
- E. Small cell carcinoma

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Item : 17 of 21 Mark



A study is conducted to characterize a familial condition in which affected individuals develop severe sunburn upon exposure to small amounts of sunlight. Examination of DNA from these individuals reveals a large number of mutations such as the one shown in the diagram. Which of the following is most likely to develop in individuals with this genetic disease?

- A. **Basal cell carcinoma**
- B. Colorectal adenocarcinoma
- C. Ductal carcinoma
- D. Hepatocellular carcinoma
- E. Small cell carcinoma

Explanation & ReKaps

The correct answer is A. The diagram demonstrates the structure of a ***thymine dimer***, which results when UV radiation

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Explanation & ReKaps

The correct answer is A. The diagram demonstrates the structure of a ***thymine dimer***, which results when UV radiation causes a covalent linkage between two adjacent thymine nucleotides on the same strand. Inability to repair this defect, also known as ***nucleotide excision repair***, results in ***xeroderma pigmentosum***, which is characterized by extreme UV sensitivity, excessive freckling, ***early onset skin cancer (basal cell, squamous cell and malignant melanoma)***, and corneal ulcerations with or without neurologic abnormalities. It is transmitted with an ***autosomal recessive*** pattern of inheritance.

None of the other answer choices are associated with mutations in nucleotide excision repair genes and UV-induced thymine dimers:

Colorectal adenocarcinoma **(choice B)** is a cancer of the ***intestinal gland cells*** that line the inside of the colon and rectum. This type of cancer ***accounts for more than 95% of colorectal cancers***, and is commonly associated with ***mutations in the tumor suppressor genes APC and TP53***, as well as ***activation of the oncogene KRAS.***

Ductal carcinoma **(choice C)** is the ***most common form of breast cancer***, arising from cells of the milk gland of the breast.These cancers ***commonly have mutations in the BRCA1 or BRCA2 genes***.

Hepatocellular carcinoma **(choice D)** is the ***most common malignancy of the liver***, frequently associated with ***chronic viral hepatitis, dietary aflatoxin B1 exposure, and alcoholism***. Mutations in ***TP53*** are common.

Small cell carcinoma **(choice E)** is a type of ***highly malignant lung cancer***. Approximately 15% of all lung cancers are small cell carcinomas, and it is ***almost exclusively found in individuals who smoke***. Mutations in several genes have been linked to this cancer, including ***TP53*** and ***RB1***.

Updated on 05/09/16

ReKaps

- Thymine dimers result when UV radiation causes a covalent linkage between two adjacent thymine nucleotides on the same strand.
- Inability to repair this defect via nucleotide excision repair results in xeroderma pigmentosum.
- Xeroderma pigmentosum is characterized by extreme UV sensitivity, excessive freckling, multiple skin cancers, and other variable manifestations depending on the locus involved. It is inherited in an autosomal recessive fashion.

Small cell carcinoma **(choice E)** is a type of ***highly malignant lung cancer***. Approximately 15% of all lung cancers are small cell carcinomas, and it is ***almost exclusively found in individuals who smoke***. Mutations in several genes have been linked to this cancer, including ***TP53*** and ***RB1***.

Updated on 05/09/16

recessive fashion.

Page References

MedEssentials (4th Ed.): pp. 64, 451

First Aid (2016): pp. 53.1, 445.1

First Aid (2015): pp. 67.1, 443.1

First Aid (2014): pp. 71.1, 438.1

First Aid (2013): pp. 69.1, 403.1

Choice A: 72% chose this answer.

Choice B: 7% chose this answer.

Choice C: 3% chose this answer.

Choice D: 16% chose this answer.

Choice E: 2% chose this answer.

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Item : 18 of 21

Mark

Previous Next

Lab Values Notes Text Zoom

A 4-month-old boy who appeared healthy at birth is brought to the physician because of poor growth and little interest in feeding. He has failed to achieve major developmental milestones, and is now in the 5th percentile for length. Physical examination shows listlessness, and distinct appearing hair that is kinky, matted, sparse, and pale in color. Laboratory studies show decreased serum ceruloplasmin and copper levels. An arteriogram shows elongation and tortuosity of major arteries. The activity of which of the following enzymes is most likely decreased in this child?

- A. Cytochrome b_5 reductase
- B. Glutathione peroxidase
- C. Lysyl oxidase
- D. N-acetylglucosamine-1-phosphotransferase
- E. Phenylalanine hydroxylase

Show Explanation

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A 4-month-old boy who appeared healthy at birth is brought to the physician because of poor growth and little interest in feeding. He has failed to achieve major developmental milestones, and is now in the 5th percentile for length. Physical examination shows listlessness, and distinct appearing hair that is kinky, matted, sparse, and pale in color. Laboratory studies show decreased serum ceruloplasmin and copper levels. An arteriogram shows elongation and tortuosity of major arteries. The activity of which of the following enzymes is most likely decreased in this child?

A. Cytochrome b_5 reductase
B. Glutathione peroxidase
C. **Lysyl oxidase**
D. N-acetylglucosamine-1-phosphotransferase
E. Phenylalanine hydroxylase

Explanation & ReKaps

The correct answer is C. The disease described is the X-linked recessive disorder, ***Menkes disease*** (also known as ***kinky hair disease***), which is characterized by kinky, depigmented, lusterless hair as well as a number of vascular, neurologic, facial, and ocular manifestations. The disease occurs due to ***mutations in the gene ATP7A, which encodes an ATP-dependent copper efflux protein in the intestine***. Copper can be absorbed into the mucosal cell, but it cannot be transported into the bloodstream. This leads to a ***severe copper deficiency*** and all copper-requiring enzymes will be adversely affected. One of these enzymes, ***lysyl oxidase***, is involved in crosslinking collagen fibers, which accounts for the connective tissue involvement and vascular abnormalities seen in this disorder. An overview of the synthesis of collagen is provided below. Additional ***copper-dependent enzymes*** affected include ***cytochrome C oxidase*** (likely leading to neurologic symptoms) and ***tyrosinase*** (likely leading to the hypopigmentation of the hair and skin).

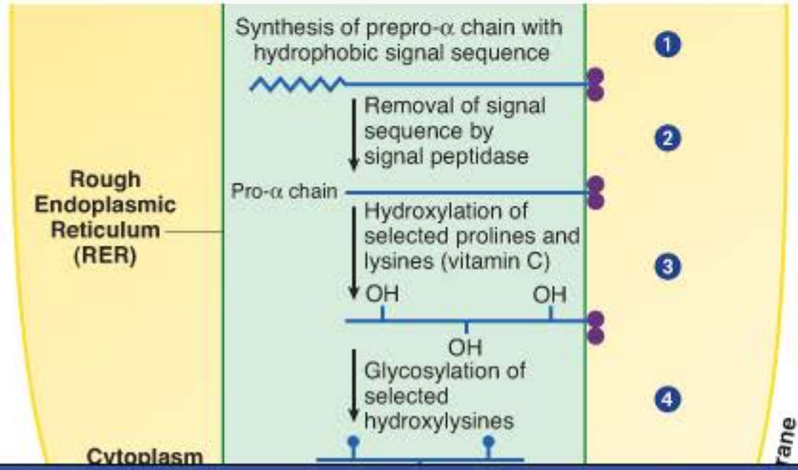
ReKaps

- Menkes disease is an X-linked recessive disorder caused by mutations in the *ATP7A* gene, resulting in an inability to transport copper into the bloodstream.
- An affected individual will have severe copper

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Explanation & ReKaps

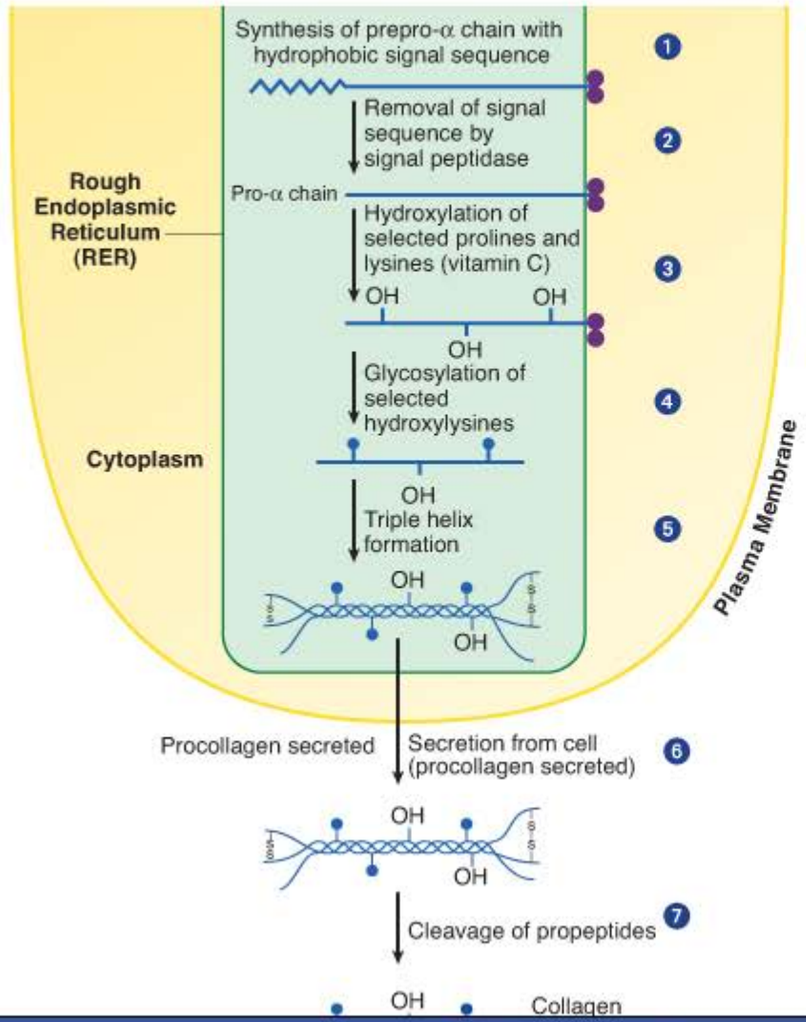
The correct answer is C. The disease described is the X-linked recessive disorder, ***Menkes disease*** (also known as ***kinky hair disease***), which is characterized by kinky, depigmented, lusterless hair as well as a number of vascular, neurologic, facial, and ocular manifestations. The disease occurs due to ***mutations in the gene ATP7A, which encodes an ATP-dependent copper efflux protein in the intestine***. Copper can be absorbed into the mucosal cell, but it cannot be transported into the bloodstream. This leads to a ***severe copper deficiency*** and all copper-requiring enzymes will be adversely affected. One of these enzymes, ***lysyl oxidase***, is involved in crosslinking collagen fibers, which accounts for the connective tissue involvement and vascular abnormalities seen in this disorder. An overview of the synthesis of collagen is provided below. Additional ***copper-dependent enzymes*** affected include ***cytochrome C oxidase*** (likely leading to neurologic symptoms) and ***tyrosinase*** (likely leading to the hypopigmentation of the hair and skin).



ReKaps

- Menkes disease is an X-linked recessive disorder caused by mutations in the *ATP7A* gene, resulting in an inability to transport copper into the bloodstream.
- An affected individual will have severe copper deficiency, and all copper-requiring enzymes (including lysyl oxidase) will be adversely affected.
- There is kinky, depigmented, lusterless hair as well as a number of vascular, neurologic, facial, and ocular manifestations.

Page References



deficiency, and all copper-requiring enzymes (including lysyl oxidase) will be adversely affected.

- There is kinky, depigmented, lusterless hair as well as a number of vascular, neurologic, facial, and ocular manifestations.

Page References

MedEssentials (4th Ed.): pp. 68

First Aid (2016): pp. 63.2

First Aid (2015): pp. 77.1

First Aid (2014): pp. 81.1

First Aid (2013): pp. 78.3

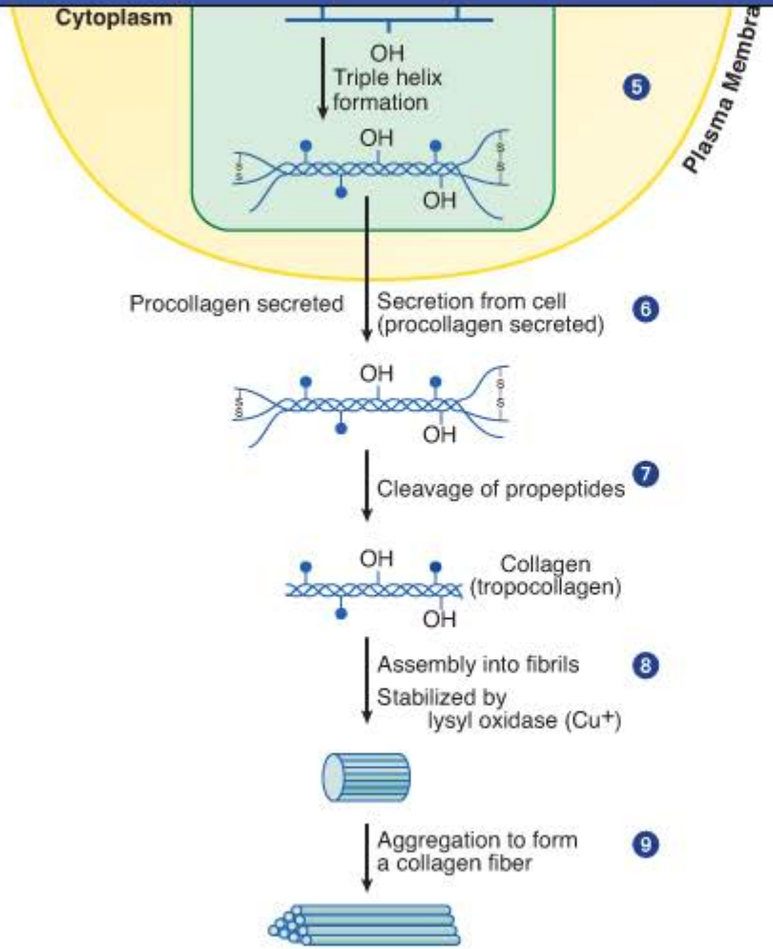
Choice A: 10% chose this answer.

Choice B: 8% chose this answer.

Choice C: 60% chose this answer.

Choice D: 10% chose this answer.

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Abnormally low activity of cytochrome b_5 reductase **(choice A),** an iron-dependent enzyme, would result in ***cyanosis and methemoglobinemia.***

MedEssentials (4th Ed.): pp. 68

First Aid (2016): pp. 63.2

First Aid (2015): pp. 77.1

First Aid (2014): pp. 81.1

First Aid (2013): pp. 78.3

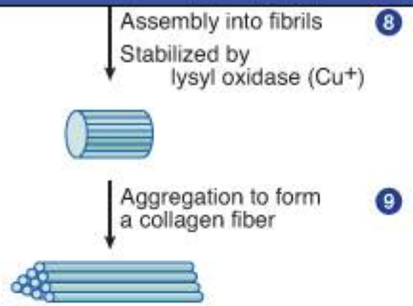
Choice A: 10% chose this answer.

Choice B: 8% chose this answer.

Choice C: 60% chose this answer.

Choice D: 10% chose this answer.

Choice E: 12% chose this answer.



Abnormally low activity of cytochrome b_5 reductase **(choice A),** an iron-dependent enzyme, would result in ***cyanosis and methemoglobinemia***.

Low levels of glutathione peroxidase activity **(choice B)** can result from ***selenium deficiency***. The clinical presentation includes myopathy and cardiomyopathy.

Deficiency of N-acetylglucosamine-1-phosphotransferase **(choice D)** results in ***I-cell disease,*** characterized by mental retardation, skeletal abnormalities, a high, narrow forehead, and coarse facies. Inclusion bodies of undigested substrates occur in cells of many tissues as well as secretion of several lysosomal enzymes into the blood and extracellular fluid. "Kinky" hair, a hallmark feature of Menkes disease, is not seen in I-cell disease.

Abnormally low phenylalanine hydroxylase activity **(choice E)** would result in ***phenylketonuria (PKU)***. Phenylalanine hydroxylase requires tetrahydrobiopterin as a coenzyme. Untreated individuals with PKU develop mental retardation after birth, but have none of the other symptoms described in this child (especially the distinct features of the patient's hair), although both Menkes patients and untreated PKU patients show hypopigmentation of the skin.

Updated on 10/11/15

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Item : 19 of 21 Mark

Previous Next

Lab Values Notes Text Zoom

The simian virus 40 (SV40) has been shown to use elements of its host replisome. In an in vitro study of SV40, investigators isolated a host enzyme that synthesized a short strand of RNA, complementary to the template strand of DNA and containing a free 3'-OH end. Which of the following enzymes is most likely involved?

- A. DNA ligase
- B. Polymerase I
- C. Polymerase III
- D. Primase
- E. Topoisomerase

Show Explanation

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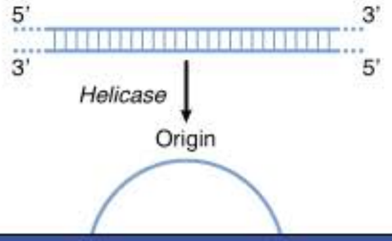
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A. DNA ligase
B. Polymerase I
C. Polymerase III
D. **Primase**
E. Topoisomerase

Explanation & ReKaps

The correct answer is D. The primer molecule required by DNA polymerase is a short strand of RNA (4-10 bases) complementary to the template strand of the DNA molecule. ***The primer is synthesized by a specific RNA polymerase known as primase.*** The growing end of the RNA primer is a free 3'-OH group. The primase does not itself require a primer for initiation of nucleotide synthesis.



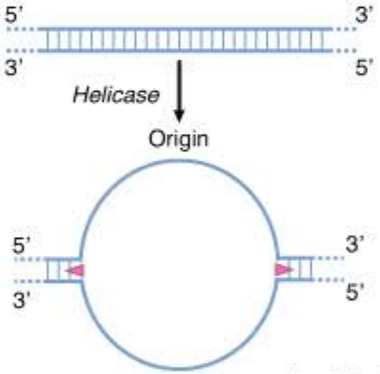
ReKaps

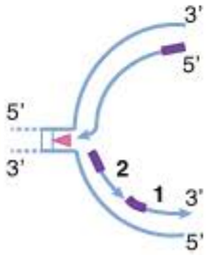
- Primase: Complex of proteins that synthesizes a short fragment of RNA. This is required to be present before the DNA can be synthesized.
- DNA polymerases: (alpha-epsilon): Read the template in the 3' to 5' direction, and

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Explanation & ReKaps

The correct answer is D. The primer molecule required by DNA polymerase is a short strand of RNA (4-10 bases) complementary to the template strand of the DNA molecule. ***The primer is synthesized by a specific RNA polymerase known as primase.*** The growing end of the RNA primer is a free 3'-OH group. The primase does not itself require a primer for initiation of nucleotide synthesis.





Leading Strand Synthesis (Continuous)

1. *Primase* synthesizes the primer (▬) 5' to 3'.
2. *DNA polymerases* α and δ extend the primer, moving **into** the replication fork (Leading strand synthesis).
3. *Helicase* (◄) continues to unwind the DNA.

Lagging Strand Synthesis (Discontinuous)

1. *Primase* synthesizes the primer (▬) 5' to 3'.
2. *DNA polymerases* α and δ extend the primer, moving **away from** the replication fork (Lagging strand synthesis).

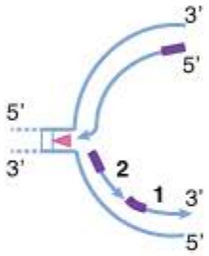
ReKaps

- Primase: Complex of proteins that synthesizes a short fragment of RNA. This is required to be present before the DNA can be synthesized.
- DNA polymerases: (alpha-epsilon): Read the template in the 3' to 5' direction, and synthesize DNA from 5' to 3'.
- DNA ligase: Joins the final phosphodiesterase bond to seal any nicks in the DNA
- DNA helicase: Uses ATP energy to unwind the dsDNA at the replication fork
- DNA topoisomerases: Relieve tension on the DNA upstream and downstream of the replication fork by cutting the DNA, allowing it to unravel, and resealing the nick.



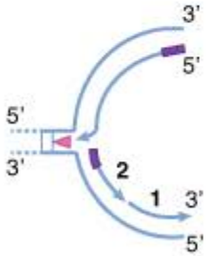
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Lagging Strand Synthesis (Discontinuous)

1. *Primase* synthesizes the primer (▬) 5' to 3'.
2. *DNA polymerases* α and δ extend the primer, moving **away from** the replication fork (Lagging strand synthesis).
3. Synthesis stops when *DNA polymerase* encounters the primer of the leading strand on the other side of the diagram (not shown), or the primer of the previous (Okasaki) fragment.
4. As *helicase* opens more of the replication fork, a third Okasaki fragment will be added.



RNAase H (5' exoribonuclease activity) digests the RNA primer from fragment 1. In the eukaryotic cell, an unidentified *DNA polymerase* extends the next fragment (2), to fill in the gap.

In prokaryotic cells *DNA polymerase 1* has both the 5' exonuclease activity to remove primers, and the *DNA polymerase* activity to extend the next fragment (2) to fill in the gap.



In both types of cells *DNA ligase* connects fragments 1 and 2 by making a phosphodiester bond.

synthesize DNA from 5' to 3'.

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- DNA topoisomerases: Relieve tension on the DNA upstream and downstream of the replication fork by cutting the DNA, allowing it to unravel, and resealing the nick.

Page References

MedEssentials (4th Ed.): pp. 63

First Aid (2016): pp. 51.1

First Aid (2015): pp. 65.1

First Aid (2014): pp. 69.1

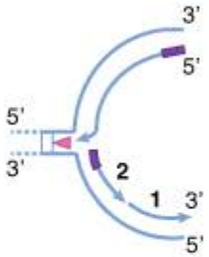
First Aid (2013): pp. 68.1

Choice A: 3% chose this answer.

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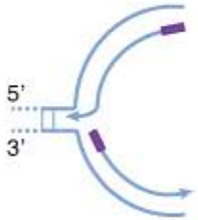
away from the replication fork (Lagging strand synthesis).

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In both types of cells *DNA ligase* connects fragments 1 and 2 by making a phosphodiester bond.

This whole process repeats to remove all RNA primers from both the leading and lagging strands.

DNA ligase **(choice A)** catalyzes the formation of a phosphodiester bond between the 3'-OH of one DNA fragment and the 5'-monophosphate group of an adjacent DNA fragment.

Polymerase I **(choice B)** catalyzes the polymerization of nucleotides and also functions in processing and repair mechanisms.

Polymerase III **(choice C)** is a part of a multiprotein complex and is the major replicating enzyme in *E. coli*.

Topoisomerases **(choice E)** produce swivel points in the DNA molecule that relieve the strain induced by the replication fork. These enzymes cut and reseal the DNA.

Page References

MedEssentials (4th Ed.): pp. 63

First Aid (2016): pp. 51.1

First Aid (2015): pp. 65.1

First Aid (2014): pp. 69.1

First Aid (2013): pp. 68.1

Choice A: 3% chose this answer.

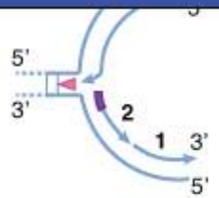
Choice B: 15% chose this answer.

Choice C: 17% chose this answer.

Choice D: 64% chose this answer.

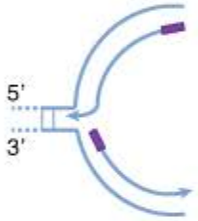
Choice E: 3% chose this answer.

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Updated on 11/02/15

MedEssentials (4th Ed.): pp. 63

First Aid (2016): pp. 51.1

First Aid (2015): pp. 65.1

First Aid (2014): pp. 69.1

First Aid (2013): pp. 68.1

Choice A: 3% chose this answer.

Choice B: 15% chose this answer.

Choice C: 17% chose this answer.

Choice D: 64% chose this answer.

Choice E: 3% chose this answer.

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Item : 20 of 21 Mark Previous Next Lab Values Notes Text Zoom

A 24-year-old woman comes to the physician because of a mole on her left forearm that seems to have increased in size over the past six months. She reports that it recently started to itch, and it bleeds when scratched. She has no family history of cancer. She works as a lifeguard during the summers and likes to surf since she was 12 years. Her temperature is 37°C (98.6°F), pulse is 72/min, respirations are 12/min, and blood pressure is 126/74 mm Hg. Physical examination shows an asymmetric mole measuring 10 mm in diameter with irregular borders and variegation. A loss of function mutation has most likely occurred in which of the following genes?

- A. *APC*
- B. *BRAF*
- C. *CDKN2A*
- D. *PTCH1*
- E. *SMAD4*

Show Explanation

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Rachel Platten
Wildfire (Track 12/12)

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Item : 20 of 21 Mark Previous Next Lab Values Notes Text Zoom

A 24-year-old woman comes to the physician because of a mole on her left forearm that seems to have increased in size over the past six months. She reports that it recently started to itch, and it bleeds when scratched. She has no family history of cancer. She works as a lifeguard during the summers and likes to surf since she was 12 years. Her temperature is 37°C (98.6°F), pulse is 72/min, respirations are 12/min, and blood pressure is 126/74 mm Hg. Physical examination shows an asymmetric mole measuring 10 mm in diameter with irregular borders and variegation. A loss of function mutation has most likely occurred in which of the following genes?

- A. *APC*
- B. *BRAF*
- C. ***CDKN2A*** ✓
- D. *PTCH1*
- E. *SMAD4*

Explanation & ReKaps

The correct answer is C. This patient's lesion has all of the clinical signs of ***melanoma***, as assessed by the "***ABCDE rule***". It is ***Asymmetrical***, its ***Borders*** are irregular, there are variations in ***Color***, its ***Diameter*** is greater than 6mm, and it is ***Evolving***. ***Mutation of CDKN2A*** is one of the most common types of mutations found in melanoma. This locus encodes two very important tumor suppressors: ***p16Ink4a*** and ***p14ARF***. The ***p16Ink4a*** protein is an ***inhibitor of cyclinD-Cdk4/6 complexes***, causing the cell to ***arrest in the G1 phase*** of the cell cycle. The ***p14ARF*** protein ***allows for the accumulation of p53,*** another tumor suppressor, by inhibiting ***MDM2***, which normally targets p53 for degradation. Mutation of *CDKN2A* results in the loss of these important tumor-suppressive activities, leading to uncontrolled proliferation of the cell. Because the question refers to a loss of function mutation (***loss of function mutation of tumor***

ReKaps

- The ABCDE rule is used to identify melanoma. It refers to Asymmetry of the growth, irregular Borders, variations in Color, a Diameter of greater than 6mm, and Evolution of the lesion

Untimed Lock

1 of 1 Playing
Superman (3:22)
Rachel Platten
Wildfire (Track 12/12)
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Explanation & ReKaps

The correct answer is C. This patient's lesion has all of the clinical signs of ***melanoma***, as assessed by the "***ABCDE rule***". It is ***Asymmetrical***, its ***Borders*** are irregular, there are variations in ***Color***, its ***Diameter*** is greater than 6mm, and it is ***Evolving***. ***Mutation of CDKN2A*** is one of the most common types of mutations found in melanoma. This locus encodes two very important tumor suppressors: ***p16Ink4a*** and ***p14ARF***. The ***p16Ink4a*** protein is an ***inhibitor of cyclinD-Cdk4/6 complexes***, causing the cell to ***arrest in the G1 phase*** of the cell cycle. The ***p14ARF*** protein ***allows for the accumulation of p53,*** another tumor suppressor, by inhibiting ***MDM2***, which normally targets p53 for degradation. Mutation of *CDKN2A* results in the loss of these important tumor-suppressive activities, leading to uncontrolled proliferation of the cell. Because the question refers to a loss of function mutation (***loss of function mutation of tumor suppressor gene***), *CDKN2A* is the correct answer.

The APC gene **(choice A)** encodes the ***adenomatous polyposis coli (APC) protein***. In healthy cells, APC is a ***tumor suppressor*** that is involved in the degradation of ***β-catenin***, preventing over proliferation of the cell. Mutation of this gene allows for the accumulation of β-catenin and abnormal cellular proliferation. Mutations in APC are associated with ***familial adenomatous polyposis (FAP)***, a ***hereditary form of colon cancer***.

BRAF **(choice B)** is a proto-oncogene encoding for a form of ***Raf kinase***, which functions in a signaling cascade to induce cell proliferation. Mutations in this gene create an ***oncogene (gain of function mutation)***, resulting in ***constitutive activation*** of this mitogenic signaling cascade. Although mutations in this gene are common in the development of melanoma, the development of melanoma, the question refers to a ***loss of function*** mutation, so this would not be an appropriate answer choice.

PTCH1 **(choice D)** is a gene that encodes for a cell surface receptor that binds to the signaling molecule ***SHH*** and determines polarity during embryonic development. It is also involved in the regulation of hair follicle formation and hair growth. It is mutated in ***nevoid basal cell carcinoma syndrome (NBCCS)***, also known as ***Gorlin syndrome***.

ReKaps

- The ABCDE rule is used to identify melanoma. It refers to Asymmetry of the growth, irregular Borders, variations in Color, a Diameter of greater than 6mm, and Evolution of the lesion over time.
- *CDKN2A* encodes for the tumor suppressors p16Ink4a and p14ARF.
- Mutations in *CDKN2A* are commonly found in melanoma and result in loss of function of these genes.

Page References

MedEssentials (4th Ed.): pp. 83, 160

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SMAD4 **(choice E)** is a member of the ***TGF β signaling*** pathway. It functions to regulate gene expression as a downstream component of this pathway. Mutations or deletions in this gene are important contributors to some forms of cancer, such as ***pancreatic adenocarcinoma***, in which more than half (55%) of all cases have lost SMAD activity.

Updated on 05/02/16

- Mutations in *CDKN2A* are commonly found in melanoma and result in loss of function of these genes.

Page References

MedEssentials (4th Ed.): pp. 83, 160

First Aid (2016): pp. 445.1

First Aid (2015): pp. 443.1

First Aid (2014): pp. 438.1

First Aid (2013): pp. 403.1

Pathoma (2015): pp. 24.4, 26 table, 208.6

Pathoma (2014): pp. 24.4, 26 table, 208.6

Pathoma (2013): pp. 24.4, 26 table, 206.6

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Item : 21 of 21 Mark Previous Next Lab Values Notes Text Zoom

A healthy 48-year-old man with a family history of colon cancer undergoes his first colonoscopy, which reveals numerous adenomatous polyps. Examination of tissue obtained on biopsy of the colon shows carcinoma. Cells in this tumor are found to have mutations in both the *KRAS* and *APC* genes. Which of the following types of genetic changes has most likely affected the *APC* gene in this patient?

- A. Gene amplification
- B. Gene deletion
- C. Promoter hyperactivity
- D. Retroviral insertion
- E. Silent mutation

Show Explanation

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

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- A. Gene amplification
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Explanation & ReKaps

The correct answer is B. ***Tumor suppressorgenes*** block tumors by regulating cell growth. Some are directly involved in ***regulation of the cell cycle*** or growth inhibition by cell-to-cell contact (***gatekeepers***). Others are involved in ***repairing DNA damage*** and ***maintaining genomic integrity*** (***caretakers***). Tumor suppressor genes must be eliminated by some mechanism such as ***deletion***, or a ***loss-of-function mutation*** in order to contribute to cancer. Familial colon cancer (***familial adenomatous polyposis***) can be due to mutations in ***APC***, a tumor suppressor gene that regulates the bifunctional protein ***catenin***. Catenin is important for cell adhesion and transcription of genes important for proliferation. *APC* promotes degradation of catenin by phosphorylation via GSK3b. ***Loss of APC*** would therefore ***promote cellular proliferation*** and contribute to the development of cancer.

There are several ways in which ***proto-oncogenes*** (normal, non-mutated) can be mutated to produce ***constitutively***

ReKaps

- Tumor suppressor genes block tumors by regulating cell growth.
- Deletion or loss-of-function mutations can lead to cancer.

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Explanation & ReKaps

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There are several ways in which ***proto-oncogenes*** (normal, non-mutated) can be mutated to produce ***constitutively active oncogenes*** (also known as ***gain-of-function mutations***). First, the gene can be amplified **(choice A)**, resulting in abnormal amounts of the protein. The gene locus can also be ***hyperactive*** due to insertion of a retrovirus **(choice D)** or increased activity of the promoter **(choice C)**.

A silent mutation **(choice E)** would not have any effect on the gene since silent mutations ***do not result in a change in a codon***. Therefore, the resulting protein's ***amino acid sequence*** will be ***unchanged***.

Updated on 10/09/15

ReKaps

- Tumor suppressor genes block tumors by regulating cell growth.
- Deletion or loss-of-function mutations can lead to cancer.
- Proto-oncogenes are normal cellular genes activated in cancer. Cancers involving these genes are usually due to gain-of-function mutations.

Page References

MedEssentials (4th Ed.): pp. 161, 358

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Updated on 10/09/15

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Page References

MedEssentials (4th Ed.): pp. 161, 358

First Aid (2016): pp. 364.1

First Aid (2015): pp. 367.1

First Aid (2014): pp. 359.1

First Aid (2013): pp. 332.1

Pathoma (2015): pp. 26.1

Pathoma (2014): pp. 26.1

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